

Learning Prices and Purchase Deadlines with Delayed and Censored Purchases

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Digital platforms increasingly pair economic offers with response windows: prices, pickup slots, promotions, and service quotes expire if customers do not act in time. This paper studies how a seller should learn prices and deadlines when the deadline itself determines which demand outcomes are observed. After observing a customer context, the seller chooses both a price and a finite purchase deadline. Customers have a latent valuation and an action-dependent consideration delay. The purchase outcome is observed only if the decision arrives before the deadline, while expired offers leave the latent buy–reject outcome censored. This creates endogenous censoring: the seller controls both the economic terms of the offer and the observation window through which demand is learned. We formulate a contextual price–deadline learning model with open-offer costs and propose *Contextual Optimization of Pricing and Deadlining with Exploration* (COPDE). The algorithm separates timing learning from demand learning, uses inverse-probability and exposure-normalized updates to recover latent demand from deadline-censored feedback, and deploys an optimistic elimination strategy to dynamically pair prices with optimal deadlines. We prove that COPDE achieves regret $\tilde{O}(T^{2/3})$ and show that this rate is optimal in T on a natural pricing subclass. The model identifies conditions under which shorter deadlines increase conversion while reducing holding costs, and shows why identical conversion rates can conceal different optimal deadlines. Numerical experiments evaluate learning performance against fixed-deadline and explore–then–commit benchmarks, examine sensitivity to the timing model, and show how higher holding costs can favor longer deadlines when lower prices accelerate customer response.

Key words: Contextual Dynamic Pricing, Time-Limited Offers, Delayed and Censored Feedback

1. Introduction

Many digital transactions now come with a clock. A ride-hailing fare may be available only for the next few minutes, a grocery pickup slot may be held for a limited time during checkout, and a flash-sale offer may expire after a specified period. In such settings, the seller is not simply choosing a price. The seller is also deciding how long the customer has to respond.

Figure 1 illustrates this design pattern in online marketplaces. The examples highlight the common operational feature that is central to our model: an economic offer is paired with a finite response window.

This second decision is easy to overlook, but it changes both customer response behavior and the seller’s learning problem in a fundamental way. Short deadlines may create urgency, release

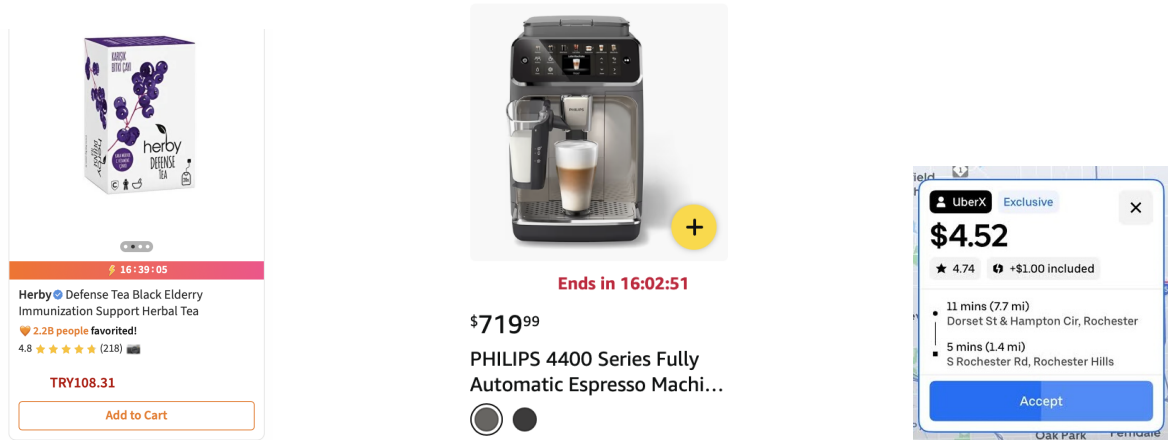


Figure 1 Illustrative examples of time-limited offers in online marketplaces. The examples emphasize the finite response window attached to the offer: the customer must respond before the offer, price, slot, or promotion expires.

scarce capacity quickly, and reduce the opportunity cost of keeping an offer open. Yet they may also cut off customers who would have purchased had they been given more time. A longer deadline gives customers more time to respond, but can also change how quickly they decide. Waiting is costly: while an offer remains active, inventory, service capacity, or promotional opportunity may be unavailable for other uses.

The difficulty is that customer responses are rarely instantaneous. After receiving an offer, a customer may compare alternatives, check availability, consult others, or simply delay the decision. If the response arrives before the deadline, the seller observes whether the customer purchased. If no response arrives by expiry, the seller cannot tell whether the customer would have purchased or rejected the offer.

This ambiguity is at the heart of our paper. Standard contextual dynamic pricing models typically assume that purchase outcomes are observed immediately after prices are posted. Standard delayed-feedback models allow outcomes to arrive late, but usually treat the delay process as external to the learner's action. Neither framework captures the setting considered here, where the seller chooses the deadline and thereby controls the observation window itself. The resulting feedback is *endogenously censored*: whether a purchase decision is observed depends directly on the seller's own price–deadline decision.

This endogeneity matters because price and deadline can affect not only revenue, but also when customers resolve the offer. Consumer research shows that time pressure can alter choice deferral, that short completion windows can create a “mere urgency” pull toward acting, and that longer

deadlines can increase perceived task difficulty and, in some settings, procrastination (Dhar and Nowlis 1999, Zhu et al. 2018, 2019). Price can shape the same resolution process: price promotions may reduce the motivation to deliberate, while anticipated regret and beliefs about future price opportunities affect whether consumers buy now or continue searching (Aydinli et al. 2014, Cooke et al. 2001, André et al. 2022). An algorithm that treats every expired offer as a rejection therefore estimates on-time conversion rather than latent demand.

We study this problem through a contextual dynamic pricing model in which, at each period, a seller observes a customer context and jointly chooses a price and a purchase deadline. The customer’s latent purchase decision follows an unknown contextual valuation model, while the time required to resolve the offer follows an action-dependent timing model. The seller earns revenue only if the customer purchases before the deadline and incurs an open-offer cost while the offer remains active. The objective therefore balances three forces: price optimization, deadline selection, and information acquisition under deadline-induced censoring.

We propose *Contextual Optimization of Pricing and Deadlining with Exploration* (COPDE), an adaptive algorithm designed for this information structure. The algorithm separates the learning problem into three components. It first explores the timing model to learn how price and deadline affect the probability that a customer’s decision arrives before expiration. Once timing estimates are sufficiently accurate, it uses censoring-adjusted price probes to learn the contextual valuation model. When both components are sufficiently accurate, it performs offset-level price elimination while choosing deadlines through a current-context best response.

A key feature of COPDE is that it does not treat each context–price–deadline combination as an unrelated arm. Instead, it pools demand information through price offsets, exploiting the single-index structure of the valuation model. Deadline effects are handled separately through the timing model, which estimates both deadline-meeting probabilities and open-offer costs. This structure allows the algorithm to learn from delayed and censored observations while preserving the statistical efficiency of contextual pricing.

Our theoretical analysis shows that COPDE achieves regret $\tilde{O}(T^{2/3})$, under standard regularity conditions. This rate is consistent with known results for contextual pricing with unknown demand and continuous prices (Luo et al. 2022, Tullii et al. 2024), while our analysis additionally accounts for learning an action-dependent timing model and for the pending observations created by deadline-censored feedback. The numerical study evaluates the cost of fixing deadlines, sensitivity to timing-model misspecification, and the consequences of ignoring action-dependent timing.

The analysis yields concrete implications for managing time-limited offers. Strong urgency can make a shorter deadline increase conversion and lower the cost of keeping an offer open at the same price. Optimal expected open time is nonincreasing in the holding-cost rate, although the announced deadline need not fall. Finally, identical conversion rates can conceal different optimal deadlines because the timing of unsuccessful offers also affects cost. Overall, the central message of our work is that deadlines in digital offers are not merely operational constraints. In fact, they are decision variables that shape both revenue and information.

Main Contributions. Our main contributions are summarized as follows.

1. **A contextual price–deadline model with endogenous censoring.** We formulate a dynamic pricing problem in which the seller jointly chooses a price and a purchase deadline after observing customer context. The deadline affects whether the customer’s delayed purchase decision is observed, making the feedback process policy-dependent.
2. **A learning framework that separates demand learning from timing learning.** We model purchase propensity through a contextual valuation index and deadline meeting through an action-dependent timing model. This structure allows the seller to estimate the latent demand curve from on-time observations while accounting for the probability that a chosen deadline censors the outcome.
3. **The COPDE algorithm.** We develop an adaptive algorithm that alternates between valuation exploration, timing exploration, and offset-level elimination with deadline best response. Demand information is pooled through permanent price-offset statistics shared across contexts, while deadlines are optimized using current estimates of deadline-meeting probabilities and open-offer costs. The algorithm uses event-driven bookkeeping, updating its estimates only when the corresponding deadline-expiry and decision-arrival events are revealed.
4. **Regret guarantees under delayed and deadline-censored feedback.** We prove that COPDE achieves regret $\tilde{O}(T^{2/3})$ under standard regularity conditions. The analysis controls three sources of learning cost: estimating contextual valuations, estimating action-dependent timing, and comparing price offsets under censored observations.
5. **Managerial implications for deadline design.** We characterize how customer response and holding costs shape deadline choice. Shortening a deadline can improve conversion while reducing open-offer costs. Optimal expected open time is nonincreasing in the holding-cost rate, without generally implying a shorter posted deadline. Our ambiguity construction further shows that purchase conversion alone need not identify the profit-maximizing deadline. These

Table 1 Representative contextual dynamic-pricing models and feedback structures.

Paper	Demand / valuation class	Unknown objects	Decision	Timing	Obs. window	Observed feedback	Regret
Filippi et al. (2010)	Known-link GLM	θ	p_t	immediate	fixed	y_t	$\tilde{O}(\sqrt{T})$
Javanmard and Nazarzadeh (2019)	Known-link linear value	α, θ	p_t	immediate	fixed	y_t	$O(\log T)$
Cohen et al. (2020)	Noiseless linear value	θ	p_t	immediate	fixed	y_t	$O(\log T)$
Ban and Keskin (2021)	Known-link heterogeneous elasticity	θ, β	p_t	immediate	fixed	y_t	$\tilde{O}(\sqrt{T})$
Chen and Gallego (2021)	Nonparametric demand surface	$q(x, p)$	p_t	immediate	fixed	y_t	$\tilde{O}(T^{\frac{d+2}{d+4}})$
Xu and Wang (2022)	Linear value / agnostic policy	value/noise	p_t	immediate	fixed	y_t	$\tilde{O}(T^{3/4})$
Luo et al. (2022, 2024)	Unknown-link linear value	θ, q	p_t	immediate	fixed	y_t	$\tilde{O}(dT^{3/4})$ $\tilde{O}(d^2T^{2/3})$
Fan et al. (2024)	Smooth unknown-link SIM	α, θ, q	p_t	immediate	fixed	y_t	$\tilde{O}((dT)^{\frac{2m+1}{4m-1}})$
Tullii et al. (2024)	Lipschitz unknown-link SIM	θ, q	p_t	immediate	fixed	y_t	$\tilde{O}((dT)^{2/3})$
Gong et al. (2026)	General value + unknown noise	v^*, F	p_t	immediate	fixed	y_t	$\tilde{O}(\rho_v^{1/3}T^{2/3})$
Wang and Chen (2025)	Smooth unknown-link SIM	α, θ, q	p_t	immediate	fixed	y_t	$\tilde{O}(T^{3/5})$ $\Omega(T^{3/5})$
This paper	Latent demand + deadline timing	θ, D, ψ	p_t, μ_t	delayed	chosen	w_t, B_t y_t if $w_t = 1$	$\tilde{O}(T^{2/3})$

Notes. SIM = single-index model. “Link” refers to the purchase-probability link in the cited pricing papers, not the timing link g_μ used in our model. d denotes the dimension of the contextual feature space. In prior contextual-pricing rows, y_t is the purchase/rejection label observed after pricing. In our row, w_t is the deadline-meeting indicator and B_t is the purchase-before-expiry indicator; the purchase label y_t is observed only if the decision arrives before the chosen deadline. The column “Obs. window” indicates whether the learner controls the window through which purchase feedback becomes observable. Regret rates suppress logarithmic and problem-dependent constants.

implications distinguish the sales, timing, and cost measures needed to evaluate time-limited offers.

The remainder of the paper is organized as follows. Section 2 reviews related work on contextual dynamic pricing, time-limited offers, and learning with delayed or censored feedback. Section 3 introduces the model and regret criterion. Section 4 presents COPDE and its event-driven learning structure. Section 5 establishes the regret guarantee. Section 6 discusses a deadline-augmented extension of the purchase model. Section 7 reports the numerical experiments. Section 8 develops the managerial implications. Section 9 concludes.

2. Literature Review

We review three key areas of research related to our setting: contextual dynamic pricing, consumer behavior under price deadlines, and bandit learning with delayed feedback.

Dynamic pricing and contextual demand learning. The closest methodological stream to our work is dynamic pricing with binary purchase feedback and contextual demand learning. In the canonical formulation, the seller observes a customer, product, or channel context, posts a price, and observes whether the customer purchases. This feedback provides limited information in the standard pricing sense: the seller observes whether the customer’s latent valuation exceeds the posted price, but not the exact valuation. Classical non-contextual models establish the basic exploration–exploitation tradeoff in pricing, including parametric demand learning, limited price experimentation, inventory-based pricing, and nonparametric or bandit formulations (Gallego and Van Ryzin 1994, Broder and Rusmevichientong 2012, Keskin and Zeevi 2014, Besbes and Zeevi 2015, Wang et al. 2014, Cheung et al. 2017, Wang et al. 2021). These papers provide the foundation for learning while pricing, but they do not model transaction-level heterogeneity through observable covariates.

A large subsequent literature incorporates such heterogeneity through contextual pricing. One stream assumes that the demand link or valuation-noise distribution is known up to finite-dimensional parameters. In this setting, generalized-linear, maximum-likelihood, semi-myopic, or feature-based methods can obtain strong regret guarantees by exploiting the known link structure (Filippi et al. 2010, Javanmard and Nazerzadeh 2019, Cohen et al. 2020, Ban and Keskin 2021). These models are powerful when the structural form of demand is reliable, but the known-link or noiseless assumption is restrictive in many pricing applications.

More recent work relaxes this structure by allowing unknown or flexible demand components. Chen and Gallego (2021) study nonparametric pricing with customer covariates, while Xu and Wang (2022) consider agnostic feature-based pricing under weaker modeling assumptions. Closest to our demand-learning component are semiparametric and distribution-free contextual-pricing models, in which purchase probabilities follow a single-index or valuation model with an unknown link or unknown valuation-noise distribution (Luo et al. 2022, 2024, Fan et al. 2024, Tullii et al. 2024, Gong et al. 2026, Wang and Chen 2025). This recent line characterizes how regret depends on smoothness, dimensionality, and structural knowledge of the demand model, and it explains why rates such as $T^{2/3}$ naturally arise when continuous prices must be optimized while an unknown demand curve is learned.

Table 1 compares these models by their demand assumptions, feedback structures, and regret guarantees.

The key distinction is that binary purchase feedback censors the customer’s valuation, whereas a deadline can censor the purchase label itself. In the cited contextual-pricing models, the observed

label directly informs demand at the posted price. Here, that inference must account for the action-dependent probability of observing the label.

Price deadlines and time-limited offers. Deadlines have long been recognized as a strategic design variable in promotions, offers, and consumer search environments. In coupon promotions, Krishna and Zhang (1999) show that expiration length affects redemption behavior, competitive interaction, and promotion profitability. More recent work studies deadlines as part of the offer itself. Hu and Tang (2021) analyze the joint choice of offer size and offer deadline when the responder searches for alternatives, showing that the optimal policy may range from an exploding offer to a longer-duration offer depending on market conditions. Gong and Huang (2025) study limited-time offers in price-directed search markets and show that such offers can be used to influence consumer search order and search prominence. Together, these papers show that deadlines are not merely implementation details. Rather, they shape how consumers search for, respond to, and evaluate offers.

A related behavioral literature explains why deadlines can affect response timing. Time pressure can change choice deferral and decision rules (Dhar and Nowlis 1999); short completion windows can create a “mere urgency” effect, inducing action even when urgency is not payoff-relevant (Zhu et al. 2018); and price promotions can reduce deliberation by making the purchase feel less consequential (Aydinli et al. 2014). Work on time-limited promotions further emphasizes that duration and discount depth interact: short deadlines may increase urgency, but their effect depends on the attractiveness of the offer and the product context (Aggarwal and Vaidyanathan 2003, Hanna et al. 2016, Eisenbeiss et al. 2015). Thus, the effect of a deadline is not necessarily monotone or universal; it operates through search, urgency, attention, and decision timing.

These papers motivate deadline choice through optimal design under known demand or search primitives, or through experimentally specified behavioral mechanisms. We study the sequential problem of learning a profitable price–deadline policy when latent demand and decision timing are unknown.

Bandits with delayed and censored feedback. A parallel line of work studies online learning when rewards are observed only after random delays. Joulani et al. (2013) provide foundational reductions for online learning under delayed feedback, showing how delay modifies regret in stochastic and adversarial settings. Subsequent work develops delayed-feedback algorithms for stochastic, linear, and generalized-linear bandits (Vernade et al. 2020, Blanchet et al. 2024, Howson et al. 2023). A particularly relevant stream is motivated by delayed conversions in online advertising

and recommendation. Chapelle (2014) models conversion delays and the bias created by treating not-yet-converted observations as negative examples, while Vernade et al. (2017) study stochastic bandits with delayed conversions, distinguishing between fully observed conversions and late conversions censored by a fixed observation window.

These studies accommodate delayed, arm-dependent, or window-censored feedback, but generally take the delay process or censoring window as part of the environment. In our setting, the seller chooses the observation window as an economic control, and the price–deadline decision can also affect the response-time distribution. The deadline enters the profit objective through both on-time purchase probabilities and holding costs, linking feedback design to offer profitability.

3. Model Formulation

We study a contextual dynamic pricing problem faced by a seller over a finite horizon of T discrete periods. At each period $t \in \{1, \dots, T\}$, a customer arrives with an observable context $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$, which may encode customer, product, or channel information. The context is observed before the seller chooses an action. We allow the context sequence to be arbitrary or stochastic, provided that x_t is nonanticipative with respect to the current-period valuation and timing shocks.

Valuation and purchase decision. The customer’s intrinsic valuation for the product is

$$u_t = x_t^\top \theta + \xi_t, \quad (1)$$

where $\theta \in \mathbb{R}^d$ is unknown and ξ_t is an i.i.d. valuation shock with distribution $\mathbb{P}_\xi$. Let F_ξ denote the CDF of ξ_t , and define the unknown demand function $D(\epsilon) := 1 - F_\xi(\epsilon)$. For a posted price p_t , the customer’s latent purchase outcome is characterized by

$$y_t = \mathbb{I}\{u_t \geq p_t\}, \quad \mathbb{E}[y_t \mid x_t, p_t] = D(p_t - x_t^\top \theta).$$

The time required to resolve the offer is modeled separately below. Thus purchase propensity follows a shifted single-index structure (e.g., Powell et al. 1989). Both the valuation parameter θ and the demand function $D(\cdot)$ are unknown and must be learned.

Price, deadline, and consideration delay. At each period t , after observing x_t , the seller jointly chooses a price $p_t \in \mathcal{P} \subseteq \mathbb{R}_+$ and a deadline duration $\mu_t \in \mathcal{M} := \{\underline{\mu}, \underline{\mu} + 1, \dots, \bar{\mu}\}$, where $\underline{\mu}$ and $\bar{\mu}$ are positive integers with $1 \leq \underline{\mu} \leq \bar{\mu} < \infty$. The offer expires at time $t + \mu_t$. We assume without loss of generality that $\mathcal{P} \subseteq [\underline{p}, \bar{p}] \subseteq [0, 1]$, because prices can be rescaled.

After receiving the offer, the customer experiences a latent consideration delay $d_t \in \mathbb{N}$. If the decision arrives by the deadline ($d_t \leq \mu_t$), the seller observes the purchase outcome. If the

decision arrives after the deadline ($d_t > \mu_t$), the offer has expired and the purchase outcome is censored. Thus the seller's deadline choice determines the observation window and makes censoring action-dependent.

Observation model and timing. Define the deadline-meeting indicator

$$w_t := \mathbb{I}\{d_t \leq \mu_t\}. \quad (2)$$

The variable w_t reveals whether the customer's decision arrived before the announced deadline. Operationally, w_t is known no later than time $t + \mu_t$: if a decision arrives before the deadline, then $w_t = 1$; if no decision arrives by the deadline, then $w_t = 0$.

If $w_t = 1$, the latent purchase label y_t is revealed at time $t + d_t \leq t + \mu_t$. If $w_t = 0$, the label y_t is not observed. We write

$$B_t := w_t y_t$$

for the purchase-before-expiry indicator. Unlike the latent label y_t , the variable B_t is known by deadline maturity: if $w_t = 1$, then $B_t = y_t$, and if $w_t = 0$, then $B_t = 0$. Hence an expired offer is not the same object as a rejection. A rejection reveals $w_t = 1$ and $y_t = 0$, whereas a censored late decision reveals $w_t = 0$ and leaves y_t missing.

Figure 2 illustrates the three possible feedback paths after a posted price–deadline offer: an observed purchase, an observed rejection, and a censored late decision.

Delay/censoring link model. Let $h_t := h(x_t, p_t, \mu_t) \in \mathbb{R}^{d'}$ be a known timing feature map, and let $\psi \in \mathbb{R}^{d'}$ be an unknown timing parameter. We assume that the deadline-meeting probability follows the link model

$$\mathbb{E}[w_t | x_t, p_t, \mu_t] = \Psi(x_t, p_t, \mu_t) = g_{\mu_t}(h_t^\top \psi), \quad (3)$$

where $\{g_\mu : \mathbb{R} \rightarrow (0, 1)\}_{\mu \in \mathcal{M}}$ is a known family of link functions.

For profit evaluation, we also assume that the expected open time has the same index structure:

$$C(x, p, \mu) := \mathbb{E}[\min\{d, \mu\} | x, p, \mu] = c_\mu(h(x, p, \mu)^\top \psi), \quad (4)$$

where $\{c_\mu\}_{\mu \in \mathcal{M}}$ is a known family of open-time functions. Section 7 evaluates the robustness of the algorithm to misspecification of the timing link.

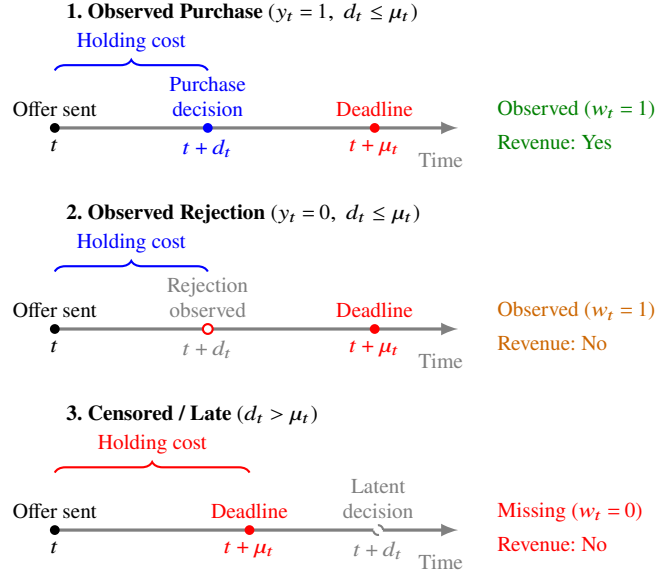


Figure 2 Operational feedback under a posted price and deadline. When the decision arrives before the deadline, the outcome is observed; otherwise it is censored. Holding costs accrue until the decision time if $d_t \leq \mu_t$, and until the deadline if $d_t > \mu_t$.

Per-round profit. The seller earns revenue only if the customer buys and the decision arrives by the deadline. The seller also pays an opportunity or holding cost proportional to the time the offer is kept open, $\min\{d_t, \mu_t\}$. Conditional on (x_t, p_t, μ_t) , the expected profit is

$$r(x_t; p_t, \mu_t) := p_t \mathbb{P}(u_t \geq p_t, d_t \leq \mu_t \mid x_t, p_t, \mu_t) - \bar{h} \mathbb{E}[\min\{d_t, \mu_t\} \mid x_t, p_t, \mu_t], \quad (5)$$

where $\bar{h} > 0$ is a known holding-cost rate. This term is a reduced-form proxy for the opportunity cost of tying up scarce capacity while the offer remains active, in the spirit of bid-price/shadow-value controls in revenue management and reservation systems in capacity-constrained services (Talluri and Van Ryzin 1998, Van Ryzin and Talluri 2005, Alexandrov and Lariviere 2012, Oh and Su 2022). For settings in which capacity is committed over time through advance reservations, see also (Chen et al. 2017, Lei and Jasin 2020).

The key structural assumption is that the buy–reject margin and the deadline-meeting process are conditionally separated given the observed offer:

$$y_t \perp\!\!\!\perp w_t \mid (x_t, p_t, \mu_t). \quad (6)$$

Equivalently, after conditioning on the context, price, and deadline, residual variation in the valuation shock is separated from residual variation in the consideration-time shock. Under this baseline separation,

$$\mathbb{P}(u_t \geq p_t, d_t \leq \mu_t \mid x_t, p_t, \mu_t) = D(p_t - x_t^\top \theta) \Psi(x_t, p_t, \mu_t), \quad (7)$$

and the expected profit factorizes as

$$r(x; p, \mu) = pD(p - x^\top \theta) \Psi(x, p, \mu) - \bar{h}C(x, p, \mu). \quad (8)$$

This factorization is the structural basis for separately learning latent demand and deadline-meeting behavior from censored observations.

Deadline choice affects realized conversion through both the time available to respond and the customer's consideration process. This formulation accommodates urgency: a shorter deadline can accelerate offer resolution and, when this acceleration outweighs the shorter response window, increase the probability of purchase before expiration. Appendix EC.7 illustrates this mechanism. In the baseline specification, these behavioral responses operate through decision timing, with intrinsic valuation held fixed across deadlines. Section 6 extends the model to allow the deadline to shift the latent buy–reject margin directly as well.

Policies and regret. Let $\mathcal{H}_t$ denote the observable history immediately before the seller chooses (p_t, μ_t) in period t . Formally, $\mathcal{H}_t$ is the sigma-field generated by past contexts and actions, together with all feedback that has matured before period t :

$$\mathcal{H}_t := \sigma(\{x_s, p_s, \mu_s : s < t\}, \{y_s : s < t, d_s \leq \mu_s, s + d_s < t\}, \{w_s, B_s : s < t, s + \mu_s < t\}). \quad (9)$$

Thus the seller may use early-arriving purchase labels as soon as they are observed and may use the deadline-maturity indicators once the corresponding deadlines have passed.

An admissible policy is a sequence of nonanticipative decision rules $\pi = \{\pi_t\}_{t=1}^T$, where each π_t is a stochastic kernel from the current context and observable history to the action space:

$$\pi_t(\cdot \mid x_t, \mathcal{H}_t) \in \Delta(\mathcal{P} \times \mathcal{M}), \quad (p_t, \mu_t) \sim \pi_t(\cdot \mid x_t, \mathcal{H}_t). \quad (10)$$

For each context x , define the oracle price–deadline pair

$$(p^*(x), \mu^*(x)) \in \arg \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} r(x; p, \mu). \quad (11)$$

The regret of policy π over horizon T is

$$\text{Regret}_T(\pi) := \sum_{t=1}^T [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)]. \quad (12)$$

ASSUMPTION 1. *We impose the following regularity and observability conditions.*

1. (Bounded operating domain) *The price set, deadline set, contexts, parameters, timing features, valuations, and timing indices are uniformly bounded. Specifically, there exist finite constants $B, B_h, B_z < \infty$, with $B_z \geq BB_h$, such that*

$$\|x\| \leq B, \quad \|\theta\| \leq B, \quad \|\psi\| \leq B, \quad \|h(x, p, \mu)\| \leq B_h$$

for all $(x, p, \mu) \in \mathcal{X} \times \mathcal{P} \times \mathcal{M}$. Hence, for every ϑ with $\|\vartheta\| \leq B$,

$$|h(x, p, \mu)^\top \vartheta| \leq B_z.$$

In addition, $\mathcal{P} \subseteq [\underline{p}, \bar{p}] \subseteq [0, 1]$, $\mathcal{M} = \{\underline{\mu}, \dots, \bar{\mu}\}$ is finite, and customer valuations satisfy

$$u_t \in [\underline{p}, \bar{p}]$$

almost surely on the operating domain.

2. (Stochastic primitives) *The valuation shocks $\{\xi_t\}$ are i.i.d. and zero-mean, and are independent of the past conditional on the current context. Conditional on (x_t, p_t, μ_t) , the timing shock that determines d_t is independent of the valuation shock ξ_t , giving the conditional separation in (6). Across periods, the valuation and timing shocks are conditionally independent given the realized contexts and actions.*
3. (Smoothness and timing information) *On the operating domain, D is L_D -Lipschitz; each g_μ is twice differentiable with*

$$|g'_\mu(z)| \leq L_g, \quad \forall \mu \in \mathcal{M}, |z| \leq B_z;$$

each c_μ is differentiable with

$$|c'_\mu(z)| \leq L_C, \quad \forall \mu \in \mathcal{M}, |z| \leq B_z;$$

and $p \mapsto h(x, p, \mu)$ is L_h -Lipschitz. In addition, for the Bernoulli timing loss

$$\ell_\mu(w, z) := -w \log g_\mu(z) - (1 - w) \log(1 - g_\mu(z)), \quad w \in \{0, 1\},$$

there exist constants $R_\ell < \infty$ and $\underline{\kappa}_\ell > 0$ such that

$$|\partial_z \ell_\mu(w, z)| \leq R_\ell, \quad \partial_{zz}^2 \ell_\mu(w, z) \geq \underline{\kappa}_\ell,$$

for all $w \in \{0, 1\}$, $\mu \in \mathcal{M}$, and $|z| \leq B_z$.

Assumption 1 is an operating-domain condition. Prices are posted within a business-defined range, deadlines are selected from a finite menu of admissible offer windows, and features are normalized before deployment. Bounded contexts, bounded parameters, and Lipschitz demand are standard in contextual dynamic pricing and ensure that nearby price offsets have comparable purchase probabilities (Javanmard and Nazerzadeh 2019, Cohen et al. 2020, Luo et al. 2022, Tullii et al. 2024). The bounded valuation condition is the corresponding normalized-price analogue: the analysis focuses on the market segment and price range in which the seller operates.

The timing conditions ensure that the deadline process is learnable on the same operating domain. Since the timing index is bounded, $\mathcal{M}$ is finite, and each g_μ is continuous with values in $(0, 1)$, deadline-meeting probabilities are bounded away from zero and one:

$$\pi_{\min} := \min_{\mu \in \mathcal{M}} \min_{|z| \leq B_z} g_\mu(z) > 0, \quad \pi_{\max} := \max_{\mu \in \mathcal{M}} \max_{|z| \leq B_z} g_\mu(z) < 1.$$

The bounded-score and loss-curvature conditions rule out locally uninformative timing likelihoods and support the self-normalized confidence bound for ψ .

REMARK 1. Our regularity conditions naturally accommodate conventional time-to-conversion models (Chapelle 2014). For example, consider a discrete consideration time $d_t = \lceil \tilde{d}_t \rceil$ generated by a latent exponential baseline model $\tilde{d}_t \mid x_t, p_t, \mu_t \sim \text{Exp}(\exp(h_t^\top \psi))$. Letting $u = h_t^\top \psi$, this yields the discrete deadline-meeting probability $g_\mu(u) = 1 - \exp(-\mu e^u)$ and expected open time $c_\mu(u) = (1 - \exp(-\mu e^u)) / (1 - \exp(-e^u))$. As we verify in Appendix EC.3, this canonical specification yields a Bernoulli timing loss with uniformly bounded score and strictly positive curvature on bounded operating domains, fully satisfying Assumption 1.

4. Proposed Algorithm

In this section we present *Contextual Optimization of Pricing and Deadlining with Exploration* (COPDE), a confidence-bound algorithm for joint price–deadline learning under delayed and deadline-censored feedback. The algorithm learns two coupled objects: a contextual valuation model governing latent purchase propensity, and a deadline-meeting model governing whether the latent decision becomes observable before expiration. A central feature of COPDE is that it does not treat censored valuation probes as missing full-observation samples. Instead, it uses the timing model to correct censored purchase observations through inverse-probability weighting.

Action discretization. The parameter estimates $\widehat{\theta}_t, \widehat{\psi}_t$ and their design matrices use feedback that has matured by calendar time t . Decisions in round t use the estimates indexed by $t - 1$. Since deadlines are integer valued, $s + \mu_s < t$ is equivalent to $s + \mu_s \leq t - 1$. For COPDE and its upper-bound analysis, the admissible price set is the full interval $\mathcal{P} = [\underline{p}, \overline{p}]$. COPDE discretizes the price dimension through an offset grid $\Delta := \{\delta_1, \dots, \delta_K\}$. Given context x_t and the current valuation estimate $\widehat{\theta}_{t-1}$, offset k induces the candidate price $p_{t,k} = x_t^\top \widehat{\theta}_{t-1} + \delta_k$. Offsets that produce prices outside the admissible range $[\underline{p}, \overline{p}]$ are discarded at the current context. Deadlines are selected from the finite menu $\mathcal{M} = \{\mu_1, \dots, \mu_L\} = \{\underline{\mu}, \underline{\mu} + 1, \dots, \bar{\mu}\}$. Thus, at each round, the candidate decision set consists of feasible offset–deadline pairs

$$(k, \ell) \in \mathcal{A}_t \times [L], \quad \mathcal{A}_t := \{k \in [K] : p_{t,k} \in [\underline{p}, \overline{p}]\}.$$

This representation separates the continuous price decision into a contextual valuation estimate and a scalar residual price offset. The offset grid is used to learn the latent demand curve, while the deadline menu is optimized through the estimated deadline-meeting and open-time models.

Algorithm overview (three modes). COPDE uses three decision modes, ordered by the type of uncertainty that must be resolved. Each round begins by processing feedback from earlier offers whose decision-arrival or deadline-expiry events have matured. The algorithm then applies the first relevant uncertainty check: it explores the timing model if deadline-meeting probabilities are still poorly estimated; otherwise it explores the valuation model if the valuation index is still uncertain; if both sources of uncertainty are sufficiently controlled, it performs offset elimination and chooses a price–deadline pair optimistically.

The timing check is performed before the valuation check because valuation exploration uses a censoring correction based on the estimated deadline-meeting probability. Given the timing estimate $\widehat{\psi}_t$, defined below, write

$$\widehat{\Psi}_t(x, p, \mu) := g_\mu \left(h(x, p, \mu)^\top \widehat{\psi}_t \right), \quad \widehat{C}_t(x, p, \mu) := c_\mu \left(h(x, p, \mu)^\top \widehat{\psi}_t \right). \quad (13)$$

For the censoring correction, define the clipped plug-in estimate

$$\widehat{\Psi}_{t-1}^c(x, p, \mu) := \text{clip} \left(\widehat{\Psi}_{t-1}(x, p, \mu), \pi_{\min}/2, 1 \right),$$

where clipping keeps inverse-probability weights uniformly bounded.

At each round, COPDE chooses among the following modes.

- **Mode 2 (timing exploration).** If the timing model is uncertain, COPDE selects a price–deadline action with large timing-feature leverage:

$$(p_t, \mu_t) \in \arg \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_{h,t-1}^{-1}}.$$

The resulting deadline-meeting indicator w_t is observed by deadline maturity and is used to update the timing parameter ψ .

- **Mode 1 (valuation exploration).** If the timing model is sufficiently accurate but the valuation estimate remains uncertain, COPDE samples

$$p_t \sim \text{Unif}[\underline{p}, \bar{p}]$$

and sets $\mu_t = \bar{\mu}$, using a fixed reference deadline for valuation probes. The probability of observing a resolved outcome at this deadline is learned through the timing model. When the probe is posted, COPDE stores the denominator

$$q_t := \widehat{\Psi}_{t-1}^c(x_t, p_t, \bar{\mu}).$$

At deadline maturity, the valuation update is formed from the observed purchase-before-expiry indicator $B_t = w_t y_t$ and the stored inverse-probability weight $1/q_t$.

- **Mode 0 (offset elimination with deadline best response).** If both the timing and valuation models are sufficiently accurate, COPDE compares feasible price offsets using confidence bounds. For each candidate offset, the algorithm evaluates deadlines using the current estimates of deadline-meeting probabilities and open-time costs, eliminates offsets that are statistically dominated, and selects an offset–deadline pair optimistically.

This structure mirrors the information decomposition in the model. The observed purchase-before-expiry signal combines latent demand with deadline meeting. Timing exploration learns the observation process; valuation exploration uses the estimated observation process to recover latent valuation information from censored feedback; and Mode 0 uses the resulting demand and timing estimates to optimize price and deadline jointly.

Event-driven feedback and bookkeeping. COPDE updates its estimators only when the relevant feedback from earlier offers has matured. For bookkeeping, let $i_s \in \{0, 1, 2\}$ denote the mode used in round s . When $i_s = 0$, let k_s and ℓ_s denote the selected offset and deadline index, so that the posted deadline is μ_{ℓ_s} . The offset statistics below are indexed by the decision round: their argument t refers to the state available before posting the round- t offer.

For each past offer s , the deadline-meeting indicator $w_s = \mathbb{I}\{d_s \leq \mu_s\}$ is known by deadline maturity. If $w_s = 1$, the purchase label y_s is revealed at time $s + d_s \leq s + \mu_s$; if $w_s = 0$, the label y_s remains unobserved. In either case, the purchase-before-expiry indicator $B_s = w_s y_s$ is known by time $s + \mu_s$. Thus a matured offer always contributes w_s and B_s , while the latent purchase label y_s is available only for uncensored observations.

The event-processing routine uses matured observations according to the mode in which the offer was generated.

First, matured Mode 2 observations update the timing model. Each such observation contributes the Bernoulli deadline-meeting outcome w_s , together with the timing feature $h_s = h(x_s, p_s, \mu_s)$ and the posted deadline μ_s , to the quasi-likelihood estimator of ψ .

Second, matured Mode 1 observations update the valuation model through an inverse-probability-weighted signal. Let

$$R_p := \bar{p} - \underline{p}.$$

When a Mode 1 probe is posted, COPDE stores the clipped deadline-meeting estimate q_s . When the probe matures, COPDE forms

$$\hat{y}_s^{\text{ipw}} := \underline{p} + R_p \frac{B_s}{q_s}.$$

The denominator q_s is fixed at the time the probe is posted and is not recomputed using later timing estimates. This keeps the valuation update adapted to the history available when the action was chosen. Importantly, the update is well-defined even when y_s is censored: if $w_s = 0$, then $B_s = 0$.

Third, matured Mode 0 observations update the offset-level demand statistics used in elimination. For each offset k , define

$$N_k(t) := \sum_{s < t: i_s=0, k_s=k} 1,$$

the number of Mode 0 offers posted with offset k before round t . This count includes both matured and pending offers. For matured Mode 0 offers, COPDE maintains the purchase-before-expiry count

$$M_k(t) := \sum_{s < t: i_s=0, k_s=k, s+\mu_s < t} B_s,$$

and the corresponding predicted deadline-meeting exposure

$$E_k(t) := \sum_{s < t: i_s=0, k_s=k, s+\mu_s < t} e_s, \quad e_s := \hat{\Psi}_{s-1}^c(x_s, p_s, \mu_s).$$

As with q_s , the exposure term e_s is stored when the offer is posted. The resulting offset-demand estimator is

$$\widehat{D}_k(t) := \left[\frac{M_k(t)}{\max\{1, E_k(t)\}} \right]_{[0,1]}. \quad (14)$$

The deadline chosen with an offer is retained for feedback processing and profit evaluation, but demand pooling is performed at the offset level. Deadline choice is handled through the current plug-in estimates of Ψ and C . Pending Mode 0 offers contribute to $N_k(t)$, but not yet to $M_k(t)$ or $E_k(t)$. The confidence widths therefore include a pending-feedback correction for offers that have been posted but have not reached deadline maturity. See Lemma 3 for further details.

Learning the valuation parameter θ . For a Mode 1 probe, suppose first that the true deadline-meeting probability is used in the IPW denominator. Conditional on x_s and p_s , the factorization in (8) gives

$$\mathbb{E} \left[\frac{B_s}{\Psi(x_s, p_s, \bar{\mu})} \middle| x_s, p_s \right] = D(p_s - x_s^\top \theta).$$

Since Mode 1 draws $p_s \sim \text{Unif}[\underline{p}, \bar{p}]$, and $u_s = x_s^\top \theta + \xi_s \in [\underline{p}, \bar{p}]$, we have

$$\begin{aligned} \mathbb{E} \left[\underline{p} + R_p \frac{B_s}{\Psi(x_s, p_s, \bar{\mu})} \middle| x_s \right] &= \underline{p} + R_p \mathbb{E}_{p_s} [D(p_s - x_s^\top \theta) \mid x_s] \\ &= \underline{p} + R_p \mathbb{E}_{\xi_s} \left[\frac{x_s^\top \theta + \xi_s - \underline{p}}{R_p} \middle| x_s \right] \\ &= x_s^\top \theta, \end{aligned}$$

where the last equality uses $\mathbb{E}[\xi_s] = 0$. With the plug-in denominator q_s , the signal is biased by an amount controlled by the current timing-estimation error. COPDE therefore performs valuation exploration only after the timing uncertainty is below the prescribed threshold.

For a matured Mode 1 probe, define

$$z_s := x_s \widetilde{y}_s^{\text{ipw}}.$$

The ridge estimator is updated using matured Mode 1 probes:

$$V_t = \lambda I_d + \sum_{s \in \mathcal{I}_t^\theta} x_s x_s^\top, \quad \widehat{\theta}_t = V_t^{-1} \sum_{s \in \mathcal{I}_t^\theta} z_s,$$

where

$$\mathcal{I}_t^\theta := \{s \leq t : i_s = 1, s + \bar{\mu} \leq t\}, \quad N_t^\theta := |\mathcal{I}_t^\theta|.$$

For the valuation-confidence bound, define the Mode 1 timing-good event

$$\mathcal{E}_\psi^\theta(\varepsilon_\psi) := \left\{ \begin{array}{l} \sup_{p \in [\underline{p}, \bar{p}]} \left| \widehat{\Psi}_{s-1}^c(x_s, p, \bar{\mu}) - \Psi(x_s, p, \bar{\mu}) \right| \leq \varepsilon_\psi \\ \text{for every Mode 1 probe } s \leq T \end{array} \right\}.$$

This uniform control applies before the randomized probe price is drawn. Set

$$R_\theta^{\text{ipw}} := \frac{R_p}{\pi_{\min}}, \quad C_{\text{ipw}} := \frac{2R_p}{\pi_{\min}}.$$

The following lemma characterizes the resulting prediction error.

LEMMA 1. *Fix $\lambda > 0$. Suppose*

$$\mathbb{P}(\mathcal{E}_\psi^\theta(\varepsilon_\psi)) \geq 1 - \zeta_\psi.$$

Then, for any $\zeta_\theta \in (0, 1)$, with probability at least $1 - \zeta_\psi - \zeta_\theta$, the following holds simultaneously for all $t = 0, \dots, T$:

$$\left| x^\top (\widehat{\theta}_t - \theta) \right| \leq \left[\beta_t^{\theta, \text{ipw}}(\zeta_\theta) + C_{\text{ipw}} \varepsilon_\psi \sqrt{N_t^\theta} \right] \|x\|_{V_t^{-1}}, \quad \forall x \in \mathcal{X},$$

where

$$\beta_t^{\theta, \text{ipw}}(\zeta_\theta) = R_\theta^{\text{ipw}} \sqrt{2 \left[d \ln \left(1 + \frac{N_t^\theta B^2}{\lambda d} \right) + \ln \frac{1}{\zeta_\theta} \right]} + \sqrt{\lambda} \|\theta\|.$$

Learning the delay parameter ψ via a Bernoulli timing likelihood. Recall from (3) that

$$\mathbb{E}[w_t \mid x_t, p_t, \mu_t] = \Psi(x_t, p_t, \mu_t) = g_{\mu_t}(h_t^\top \psi), \quad h_t = h(x_t, p_t, \mu_t).$$

Timing exploration uses the always-matured deadline-meeting indicator w_t . In Mode 2, COPDE selects a price–deadline action with large timing-feature leverage. Once the corresponding deadline matures, the observation w_s contributes a Bernoulli likelihood term for estimating ψ . Since w_s is revealed by time $s + \mu_s$, timing updates do not require observing the latent purchase label y_s .

Let

$$\mathcal{I}_t^\psi := \{s \leq t : i_s = 2, s + \mu_s \leq t\}$$

denote the set of Mode 2 timing-exploration rounds whose deadline feedback has matured by calendar time t , and let $N_t^\psi := |\mathcal{I}_t^\psi|$. Using these observations, COPDE computes the regularized quasi-maximum likelihood estimator

$$\widehat{\psi}_t \in \arg \min_{\vartheta \in \Theta_\psi} \left\{ \frac{\lambda}{2} \|\vartheta\|_2^2 + \sum_{s \in \mathcal{I}_t^\psi} \ell_{\mu_s}(w_s, h_s^\top \vartheta) \right\},$$

where

$$\Theta_\psi := \{\vartheta \in \mathbb{R}^{d'} : \|\vartheta\|_2 \leq B\},$$

and

$$\ell_\mu(w, z) := -w \log g_\mu(z) - (1 - w) \log(1 - g_\mu(z)).$$

The corresponding timing design matrix is

$$V_{h,t} := \lambda I_{d'} + \underline{\kappa}_\ell \sum_{s \in \mathcal{I}_t^\psi} h_s h_s^\top,$$

where $\underline{\kappa}_\ell$ is the uniform curvature constant of the timing loss on the bounded operating domain.

The plug-in estimates in (13) use this timing estimator. Lemma 2 provides high-probability bounds for both the deadline-meeting probability and the expected open time.

LEMMA 2. Fix $\lambda > 0$. Suppose that, on the operating domain,

$$|\partial_z \ell_\mu(w, z)| \leq R_\ell, \quad \partial_{zz}^2 \ell_\mu(w, z) \geq \underline{\kappa}_\ell > 0$$

for all $w \in \{0, 1\}$, $\mu \in \mathcal{M}$, and $|z| \leq B_z$. Let $\mathcal{I}_t^\psi$, N_t^ψ , and $V_{h,t}$ be the matured sample set, sample count, and timing design matrix defined above. Then for any $\zeta \in (0, 1)$, with probability at least $1 - \zeta$, simultaneously for all $t = 0, \dots, T$ and all $(x, p, \mu) \in \mathcal{X} \times \mathcal{P} \times \mathcal{M}$,

$$\left| h(x, p, \mu)^\top (\widehat{\psi}_t - \psi) \right| \leq \beta_t^\psi(\zeta) \|h(x, p, \mu)\|_{V_{h,t}^{-1}}.$$

Consequently,

$$\left| \widehat{\Psi}_t(x, p, \mu) - \Psi(x, p, \mu) \right| \leq L_g \beta_t^\psi(\zeta) \|h(x, p, \mu)\|_{V_{h,t}^{-1}},$$

and

$$\left| \widehat{C}_t(x, p, \mu) - C(x, p, \mu) \right| \leq L_C \beta_t^\psi(\zeta) \|h(x, p, \mu)\|_{V_{h,t}^{-1}}.$$

Here one may take

$$\beta_t^\psi(\zeta) := \frac{R_\ell}{\sqrt{\underline{\kappa}_\ell}} \sqrt{2 \left[d' \ln \left(1 + \frac{N_t^\psi \underline{\kappa}_\ell B_h^2}{\lambda d'} \right) + \ln \frac{T}{\zeta} \right]} + \sqrt{\lambda} \|\psi\|.$$

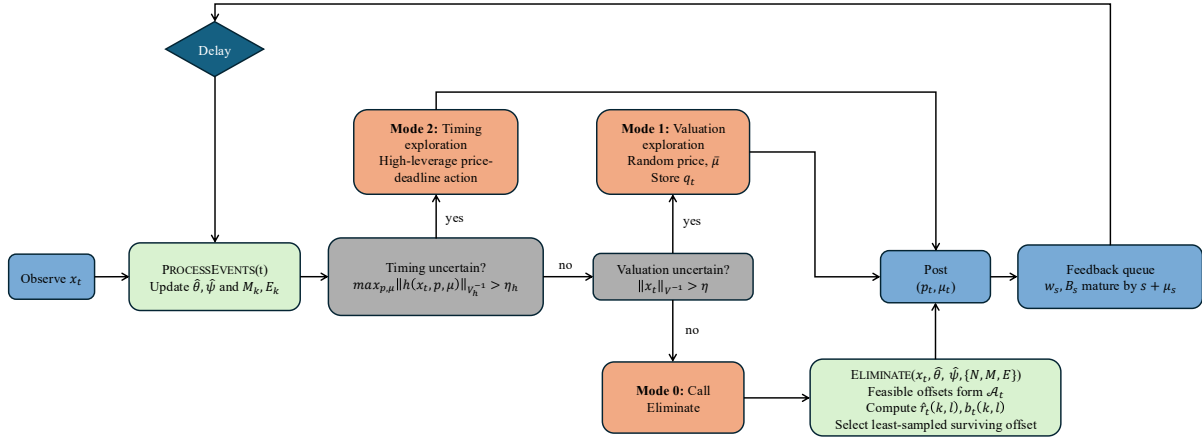


Figure 3 Schematic diagram of COPDE. Each round first processes matured feedback through **PROCESSEVENTS**. The algorithm then checks timing uncertainty before valuation uncertainty, since valuation exploration uses an inverse-probability correction based on the estimated deadline-meeting probability. If both uncertainty gates are passed, **ELIMINATE** computes current-context offset confidence bounds and returns the surviving offset set $\mathcal{K}_t$ and pairwise UCB values. The main loop then selects the least-sampled surviving offset and an optimistic deadline, posts the resulting price–deadline pair, and stores the offer in the feedback queue until the corresponding deadline-expiry or decision-arrival events mature.

Successive elimination on the offset grid. When both uncertainty gates are passed, COPDE uses the accumulated offset-level statistics to compare feasible price offsets at the current context. The elimination step exploits the single-index demand structure. If the valuation index were known exactly, posting price $x_t^\top \theta + \delta_k$ would generate latent demand $D(\delta_k)$, independent of the context. With the estimated index $x_t^\top \hat{\theta}_{t-1}$, this pooling remains valid up to the valuation-estimation error controlled by Lemma 1. Deadlines are handled separately through the timing model, because they affect realized conversion and open-offer cost through Ψ and C , rather than defining separate demand curves.

At a Mode 0 round t , COPDE evaluates the feasible candidate prices $p_{t,k}$, $k \in \mathcal{A}_t$, defined above. For each deadline μ_ℓ , COPDE computes the plug-in profit estimate

$$\hat{r}_t(k, \ell) := p_{t,k} \hat{D}_k(t) \hat{\Psi}_{t-1}(x_t, p_{t,k}, \mu_\ell) - \bar{h} \hat{C}_{t-1}(x_t, p_{t,k}, \mu_\ell), \quad (15)$$

where $\hat{D}_k(t)$ is the exposure-normalized estimator in (14). Let $\varepsilon_{\theta, \text{eff}}$ denote the valuation prediction-error level obtained from the IPW valuation estimator, and let ε_ψ denote the timing-error level. COPDE uses the confidence radius

$$b_t(k, \ell) := \kappa_2 \hat{\Psi}_{t-1}(x_t, p_{t,k}, \mu_\ell) \sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_k(t)\}}} + \kappa_5 \frac{\bar{\mu}}{\max\{1, N_k(t)\}}$$

$$+ \kappa_3 \varepsilon_{\theta, \text{eff}} + \kappa_4 \varepsilon_{\psi}. \quad (16)$$

The pairwise confidence bounds are

$$\text{UCB}_t(k, \ell) := \widehat{r}_t(k, \ell) + b_t(k, \ell), \quad \text{LCB}_t(k, \ell) := \widehat{r}_t(k, \ell) - b_t(k, \ell).$$

Because the deadline is optimized at the current context, the confidence bounds for an offset aggregate over its best deadline:

$$\text{UCB}_t(k) := \max_{\ell \in [L]} \text{UCB}_t(k, \ell), \quad \text{LCB}_t(k) := \max_{\ell \in [L]} \text{LCB}_t(k, \ell).$$

COPDE retains offsets whose optimistic value is not dominated by the pessimistic value of another feasible offset:

$$\mathcal{K}_t := \left\{ k \in \mathcal{A}_t : \text{UCB}_t(k) \geq \max_{j \in \mathcal{A}_t} \text{LCB}_t(j) \right\}.$$

It then selects the least-sampled surviving offset,

$$k_t \in \arg \min_{k \in \mathcal{K}_t} N_k(t),$$

and chooses an optimistic deadline for that offset,

$$\ell_t \in \arg \max_{\ell \in [L]} \text{UCB}_t(k_t, \ell).$$

The posted action is

$$p_t = p_{t, k_t}, \quad \mu_t = \mu_{\ell_t}.$$

LEMMA 3. *Let $p_{t, k}$ be a feasible Mode 0 candidate price at round t , let $\mu_\ell \in \mathcal{M}$, and let $\widehat{r}_t(k, \ell)$ be the profit estimator in (15). Suppose $\varepsilon_\psi \leq \pi_{\min}/2$. Then there exist constants $\kappa_2, \kappa_3, \kappa_4, \kappa_5 < \infty$, depending only on the operating-domain constants, such that for any $\zeta \in (0, 1)$, with probability at least $1 - \zeta$, the following holds simultaneously for all Mode 0 rounds t , all feasible offsets $k \in \mathcal{A}_t$, and all deadlines $\ell \in [L]$:*

$$\begin{aligned} & \left| \widehat{r}_t(k, \ell) - r(x_t; p_{t, k}, \mu_\ell) \right| \\ & \leq \kappa_2 \widehat{\Psi}_{t-1}(x_t, p_{t, k}, \mu_\ell) \sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_k(t)\}}} + \kappa_5 \frac{\bar{\mu}}{\max\{1, N_k(t)\}} \\ & \quad + \kappa_3 \varepsilon_{\theta, \text{eff}} + \kappa_4 \varepsilon_{\psi}. \end{aligned}$$

Proof sketch of Lemma 3. The estimator error has three components: the concentration error of the exposure-normalized offset estimator $\widehat{D}_k(t)$, the valuation-centering error from using $x^\top \widehat{\theta}$ rather than $x^\top \theta$, and the plug-in timing error from estimating Ψ and C . The exposure clock $E_k(t)$ appears because a matured Mode 0 observation with offset k has conditional mean approximately $D(\delta_k)\Psi(x_s, p_s, \mu_s)$. Normalizing by the stored predicted exposure e_s therefore recovers the latent demand value $D(\delta_k)$, up to concentration and plug-in errors. The pending term accounts for offers that have been posted but have not yet reached deadline maturity. A full derivation is provided in the appendix. $\square$

Algorithm 1 summarizes the main COPDE loop, and Figure 3 illustrates its decision flow. The calls to `PROCESSEVENTS` and `ELIMINATE` refer, respectively, to the event-driven update routine and the current-context offset-elimination routine. The event-processing routine uses matured Mode 2 observations to update the timing model, matured Mode 1 observations to update the valuation model through the stored IPW denominator q_s , and matured Mode 0 observations to update the offset-level statistics M_k and E_k . Detailed pseudocode for these routines is provided in Appendix EC.1.

Computational complexity. Each round performs event-driven updates and, in Mode 0, a scan over the feasible offset–deadline grid. The Mode 0 scan evaluates $\widehat{r}_t(k, \ell)$, UCB/LCB values, and offset-level maxima over $k \in \mathcal{A}_t$ and $\ell \in [L]$, giving an $O(|\mathcal{A}_t|L) \leq O(KL)$ grid cost. Event processing updates $V \in \mathbb{R}^{d \times d}$, $V_h \in \mathbb{R}^{d' \times d'}$, and the offset statistics when feedback matures. The timing update requires solving the regularized timing likelihood over matured Mode 2 observations; this numerical optimization cost is separate from the $O(KL)$ grid scan. If the timing-exploration maximization over $\mathcal{P} \times \mathcal{M}$ is evaluated on the same offset–deadline grid, its search cost is also $O(KL)$.

5. Regret Analysis

We now present the main theoretical guarantee for COPDE. The result shows that the algorithm achieves sublinear regret despite delayed feedback, deadline-induced censoring, and the need to learn both latent demand and deadline-meeting behavior.

THEOREM 1. *Let $T \geq 4$, and let $d' := \dim(h)$. Suppose the offset grid has $K = O(T^{1/3})$ points and mesh $\eta_\Delta \leq \kappa_\Delta T^{-1/3}$ over the operating offset interval. Define*

$$\varepsilon_\theta := d^{2/3} \sqrt{\log T} T^{-1/3}, \quad \varepsilon_\psi := (d')^{2/3} \sqrt{\log T} T^{-1/3}.$$

Let

$$\varepsilon_{\theta, \text{eff}} := \varepsilon_\theta + C_{\theta\psi} \sqrt{d \log T} \varepsilon_\psi,$$

Algorithm 1: COPDE: Contextual Optimization of Pricing and Deadlining with Exploration

Input: offsets $\{\delta_k\}_{k=1}^K$, deadlines $\{\mu_\ell\}_{\ell=1}^L = \mathcal{M}$, thresholds η, η_h , ridge parameter λ .

Initialize: $V \leftarrow \lambda I_d$, $b \leftarrow 0$, $\hat{\theta} \leftarrow 0$, $V_h \leftarrow \lambda I_{d'}$, $\hat{\psi} \leftarrow 0$.

For all offsets k , set $N_k \leftarrow 0$, $M_k \leftarrow 0$, $E_k \leftarrow 0$.

for $t = 1, \dots, T$ **do**

 Observe context x_t .

$(V, b, \hat{\theta}, V_h, \hat{\psi}, \{M_k, E_k\}) \leftarrow \text{PROCESSEVENTS}(t)$.

if $\max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_h^{-1}} > \eta_h$ **then**

 Choose

$$(p_t, \mu_t) \in \arg \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_h^{-1}}.$$

 Set $i_t \leftarrow 2$.

else if $\|x_t\|_{V^{-1}} > \eta$ **then**

 Sample $p_t \sim \text{Unif}[\underline{p}, \bar{p}]$.

 Set $\mu_t \leftarrow \bar{\mu}$ and $i_t \leftarrow 1$.

 Store $q_t \leftarrow \hat{\Psi}_{t-1}^c(x_t, p_t, \bar{\mu})$.

else

$(\mathcal{K}_t, \{\text{UCB}_t(k, \ell)\}_{k, \ell}) \leftarrow \text{ELIMINATE}(x_t, \hat{\theta}, \hat{\psi}, \{N, M, E\})$.

 Choose $k^* \in \arg \min_{k \in \mathcal{K}_t} N_k$.

 Choose $\ell^* \in \arg \max_{\ell \in [L]} \text{UCB}_t(k^*, \ell)$.

 Set $p_t \leftarrow x_t^\top \hat{\theta} + \delta_{k^*}$, $\mu_t \leftarrow \mu_{\ell^*}$, $i_t \leftarrow 0$. Store $(k_t, \ell_t) \leftarrow (k^*, \ell^*)$.

 Store $e_t \leftarrow \hat{\Psi}_{t-1}^c(x_t, p_t, \mu_t)$. Set $N_{k^*} \leftarrow N_{k^*} + 1$.

end

 Post (p_t, μ_t) .

end

where $C_{\theta\psi}$ is a constant depending only on the operating-domain constants and on $\pi_{\min}^{-1}$. Let $\zeta_T := 1/(6T)$, and choose

$$\eta := \frac{\varepsilon_\theta}{\beta_T^{\theta, \text{ipw}}(\zeta_T)}, \quad \eta_h := \frac{\varepsilon_\psi}{\max\{L_g, L_C\} \beta_T^\psi(\zeta_T)}.$$

Assume T is large enough that $\varepsilon_\psi \leq \pi_{\min}/2$. Under Assumption 1, with probability at least $1 - \mathcal{O}(T^{-1})$,

$$\text{Regret}_T(\pi) = \tilde{\mathcal{O}}(T^{2/3}),$$

where $\tilde{\mathcal{O}}$ hides logarithmic factors in T and problem-dependent constants.

Proof sketch of Theorem 1. The proof decomposes regret according to the three operating modes of COPDE. Because valuation exploration uses an inverse-probability correction based on the timing model, timing uncertainty is resolved first. Define

$$G_2 := \left\{ t \leq T : \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_{h, t-1}^{-1}} > \eta_h \right\},$$

and

$$G_1 := \left\{ t \leq T : t \notin G_2, \|x_t\|_{V_{t-1}^{-1}} > \eta \right\}.$$

The set G_2 contains timing-exploration rounds, and G_1 contains valuation-exploration rounds. By the elliptical-potential counting argument in Lemma EC.1,

$$|G_2| = \tilde{O}\left(\frac{d'}{\eta_h^2}\right), \quad |G_1| = \tilde{O}\left(\frac{d}{\eta^2}\right).$$

With the choices of η and η_h , the total exploration regret is therefore $\tilde{O}(T^{2/3})$.

For rounds outside $G_1 \cup G_2$, both uncertainty gates have passed. Lemma 2 implies uniform timing error at most ε_ψ . Lemma 1 and the valuation gate imply the effective valuation prediction bound

$$|x_t^\top (\widehat{\theta}_{t-1} - \theta)| \leq \varepsilon_{\theta, \text{eff}}.$$

The regret on Mode 0 rounds is split into discretization error and offset-elimination error. The discretization term is controlled by the price-offset mesh and the valuation-centering error, giving a contribution of order

$$O(T\eta_\Delta + T\varepsilon_{\theta, \text{eff}}).$$

For the elimination term, define the current candidate price

$$p_{t,k} := x_t^\top \widehat{\theta}_{t-1} + \delta_k$$

and the current grid value

$$\bar{v}_t(k) := \max_{\ell \in [L]} r(x_t; p_{t,k}, \mu_\ell).$$

Let $k_t^* \in \arg \max_{k \in \mathcal{A}_t} \bar{v}_t(k)$. On the confidence event of Lemma 3, the oracle grid offset k_t^* is not eliminated. If COPDE selects $k_t \in \arg \min_{k \in \mathcal{K}_t} N_k(t)$, then the offset-optimism argument gives

$$\bar{v}_t(k_t^*) - r(x_t; p_t, \mu_t) \lesssim \widehat{\Psi}_{t-1}(x_t, p_{t,k_t}, \mu_{\ell_t}) \sqrt{\frac{\log(KLT/\zeta_T)}{\max\{1, E_{k_t}(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_{k_t}(t)\}} + \varepsilon_{\theta, \text{eff}} + \varepsilon_\psi.$$

The permanent offset statistics are crucial here. Changing the context changes which offsets are currently feasible and competitive, but it does not reset what has been learned about $D(\delta_k)$.

Peeling over the offset counts yields

$$\tilde{O}(K/\alpha + T\alpha + T\varepsilon_{\theta, \text{eff}} + T\varepsilon_\psi + K\bar{\mu})$$

for any peeling level $\alpha > 0$. Taking $\alpha = T^{-1/3}$ and $K = O(T^{1/3})$ gives $\tilde{O}(T^{2/3})$. Combining exploration, discretization, and offset-elimination terms proves the theorem. $\square$

REMARK 2. The upper bound reveals a tradeoff in allocating exploration between timing and valuation. A tighter timing threshold increases the exploration-cost term, which scales as η_h^{-2} , while reducing both the direct timing error in profit evaluation and the error propagated through the IPW valuation update. The latter contributes $C_{\theta\psi}\sqrt{d\log T}\varepsilon_\psi$ to the valuation-error bound and is governed by timing accuracy rather than the valuation target ε_θ . Timing exploration therefore serves two purposes: improving offer comparisons and improving the censoring correction used to learn demand.

5.1. Rate optimality and deadline identification

The following results establish a fundamental limit on learning performance and explain why timing feedback matters for deadline choice. First, the $T^{2/3}$ dependence on the horizon is unavoidable in the worst case, even with a fixed context and a known timing process. Second, even exact knowledge of conversion probabilities at every admissible deadline can leave the profit-maximizing deadline undetermined. Together, these results show that COPDE attains the optimal regret rate up to logarithmic factors while learning information essential to profitable deadline decisions.

THEOREM 2. *There exists a constant $c > 0$ such that, for every horizon T and every admissible policy π ,*

$$\sup_{T \in \mathfrak{S}} \mathbb{E}_T [\text{Regret}_T(\pi)] \geq cT^{2/3},$$

where $\mathfrak{S}$ is a subclass of the price–deadline problem satisfying Assumption 1. Consequently, the $\tilde{O}(T^{2/3})$ upper bound of Theorem 1 is rate-optimal in T , up to logarithmic factors, over this class.

The construction uses a fixed context, a single deadline, and a known timing process. Holding costs are constant across prices, so regret is a fixed multiple of the regret from learning an unknown demand curve. This establishes a pricing benchmark that COPDE attains, up to logarithmic factors, while also learning action-dependent timing and choosing deadlines.

The next result shows that conversion alone need not identify the optimal deadline, even when the price is fixed.

PROPOSITION 1. *There exist two instances $\mathcal{I}_+$ and $\mathcal{I}_-$ satisfying Assumption 1, with the same fixed context, price, holding-cost rate, and known timing and open-time links, and with admissible deadlines $\mathcal{M} = \{1, 2\}$, such that*

$$\mathbb{E}_+[B_t \mid \mu_t = \mu] = \mathbb{E}_-[B_t \mid \mu_t = \mu] \quad \text{for every } \mu \in \mathcal{M},$$

yet their unique profit-maximizing deadlines differ.

The two instances generate identical expected sales revenue at each deadline, but their expected holding costs differ. Even exact knowledge of purchase-before-expiry probabilities at every admissible deadline would therefore leave the optimal deadline undetermined. The deadline-meeting feedback w_t distinguishes the two instances by revealing whether offers resolve before expiry, including those that end without a purchase. Proofs of both results are in Appendix EC.5.

6. Deadline-Augmented Purchase Propensity

In the baseline model, deadlines shape realized purchases by influencing whether and when customers resolve an offer. They may also affect the decision to buy itself, for example by changing the willingness to buy or opportunity to compare alternatives. These direct purchase effects create an additional learning challenge: the seller must learn how demand shifts with the deadline while accounting for delayed and censored feedback. We model these effects as shifts in a common demand curve and show how COPDE can learn both the curve and the deadline effects using observations from different contexts and deadlines.

Let $a(x, \mu) \in \mathbb{R}^q$ be a known feature map that captures direct deadline effects, including interactions with customer characteristics. Define $\phi(x, \mu) = (x^\top, a(x, \mu)^\top)^\top$ and $\omega = (\theta^\top, \gamma^\top)^\top$. The augmented purchase model is

$$u_t^\dagger = \phi(x_t, \mu_t)^\top \omega + \xi_t,$$

$$\Pr(y_t = 1 \mid x_t, p_t, \mu_t) = D(p_t - \phi(x_t, \mu_t)^\top \omega).$$

The unknown parameter γ captures the direct effect of the deadline on the buy–reject margin; $\gamma = 0$ recovers the baseline model. Retaining the timing model and the conditional separation of valuation and timing shocks, expected profit becomes

$$r^\dagger(x; p, \mu) = pD(p - \phi(x, \mu)^\top \omega) \Psi(x, p, \mu) - \bar{h}C(x, p, \mu). \quad (17)$$

The extension requires two changes to COPDE. First, each offset k now generates a deadline-specific candidate price,

$$p_{t,k,\ell} = \phi(x_t, \mu_\ell)^\top \widehat{\omega}_{t-1} + \delta_k.$$

The algorithm therefore compares feasible offset–deadline pairs using these candidate prices. Demand observations can still be pooled by offset:

$$p_{t,k,\ell} - \phi(x_t, \mu_\ell)^\top \omega = \delta_k + \phi(x_t, \mu_\ell)^\top (\widehat{\omega}_{t-1} - \omega).$$

Once the augmented index is estimated accurately, observations assigned to offset k inform the same demand value $D(\delta_k)$, including when they come from different deadlines.

Second, valuation exploration must learn which deadlines shift purchase propensity. Mode 1 therefore chooses a deadline for which the augmented valuation index is most uncertain, then draws a randomized price and uses the same censoring correction to construct a valuation observation. The algorithm proceeds to Mode 0 only when the valuation estimates are sufficiently accurate for every available deadline. Timing exploration continues to precede valuation exploration so that this correction uses an adequately estimated deadline-meeting probability.

Under the augmented regularity and offset-coverage conditions stated in Appendix EC.6, this version of COPDE retains regret $\widetilde{O}(T^{2/3})$. The valuation dimension becomes $d + q$, while the timing dimension remains $d' = \dim(h)$. The appendix gives the modified estimators, exploration rule, and the adaptations of the regret argument. The shared demand function D is central to this result: deadline effects shift its index, allowing demand information to be shared across offers.

7. Numerical Experiments

Our numerical study examines when learning to choose both prices and deadlines improves profit. We first compare COPDE with policies that keep deadlines fixed or stop learning after an initial exploration phase. We then examine how higher holding costs affect policy performance and the seller’s choice of offers. Finally, we test what happens when the learner ignores how its offers affect response speed or assumes the wrong response-time distribution.

Simulation setting and performance measure. Customer valuations depend on observed customer characteristics. Response times are exponential, with a rate that depends on those characteristics, the posted price, and the deadline. An offer remains open until the customer responds or the deadline expires, incurring holding costs during that time. It earns revenue only if the customer purchases before expiration. Once the deadline has passed, every learner observes whether the customer responded in time and, if so, whether the customer purchased or rejected the offer. Appendix EC.2 gives the simulation setup and implementation details.

Relative regret measures the gap between the policy’s cumulative expected profit and the oracle’s, expressed as a percentage of the oracle’s profit. Lower values indicate better performance. The

oracle knows customer demand and response timing and chooses from the full feasible price range and the same available deadlines. The comparison includes revenue and holding costs from all offers, including those made during exploration. Shaded bands and error bars show 95% confidence intervals across runs.

Compared policies. COPDE updates its estimates of customer valuations and response timing as feedback becomes available and uses them to choose a price and a deadline for each arriving customer. The *Fixed-Deadline* policies learn prices but keep a prespecified Short, Medium, or Long deadline throughout. Short gives customers less time to respond and limits how long offers remain open. Long allows more time to respond but can incur greater holding costs. *Explore–Then–Commit* (ETC) first makes exploratory offers to estimate demand and response timing. It then stops updating these estimates, while still choosing a price and deadline for each new customer using the fitted models. Appendix EC.2 describes the estimators and settings.

Learning performance. Figure 4 (Left) shows that COPDE’s relative regret declines as it learns and is the lowest among the compared policies by the end of the horizon. Medium performs best among the fixed deadlines, balancing time to respond against holding costs. COPDE achieves higher profit than Medium, showing an advantage even over a well-performing fixed deadline. ETC’s initial exploration followed by fixed estimates also leaves a larger profit shortfall. We next examine whether the same fixed deadline remains a good choice as holding costs increase.

Learning under higher holding costs. We increase the holding-cost rate, keeping customer demand, the response-time model, the available prices and deadlines, and the learning settings unchanged. Figure 4 (Right) compares policy performance across low, medium, and high holding costs.

The preferred fixed deadline changes with holding costs. Medium performs best among fixed deadlines at the low cost, whereas Short performs best at the higher costs. Long becomes particularly costly as the holding cost increases because its sales revenue no longer covers its holding costs. COPDE maintains the lowest relative regret across the tested cost levels, and its advantage over the best fixed deadline grows as holding costs increase.

How holding costs change offer decisions. We next examine how holding costs change profit-maximizing offers in a separate experiment where demand and response timing are known. We jointly optimize prices and deadlines in two scenarios with the same demand and available offers. At a common reference price, customer response rates are also the same. At the longest deadline, cutting the price speeds up response less strongly in Scenario A and more strongly in Scenario B. Within each scenario, we vary only the holding-cost rate. Appendix EC.2.4 gives the specifications.

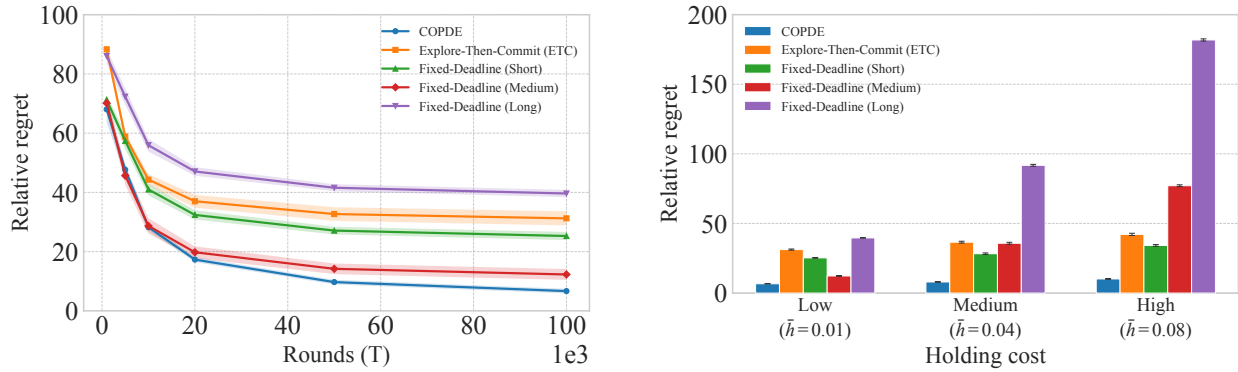


Figure 4 Learning performance and holding costs. Left: Relative regret over the learning horizon. Right: Relative regret under low, medium, and high holding costs.

Figure 5 shows that higher holding costs can lead to opposite deadline adjustments. Comparing the low- and high-cost settings, the optimal offer in Scenario A has a shorter deadline. In Scenario B, it combines a lower price with a longer deadline. The average time until the customer responds or the offer expires (expected open time) decreases in both scenarios. In Scenario A, the shorter deadline limits how long offers remain open. In Scenario B, the price reduction induces faster responses under the longer deadline, reducing expected open time despite the later expiration. A longer announced deadline therefore need not mean that offers remain open longer on average. These results illustrate why the response to higher holding costs depends on how price and deadline jointly affect customer response timing.

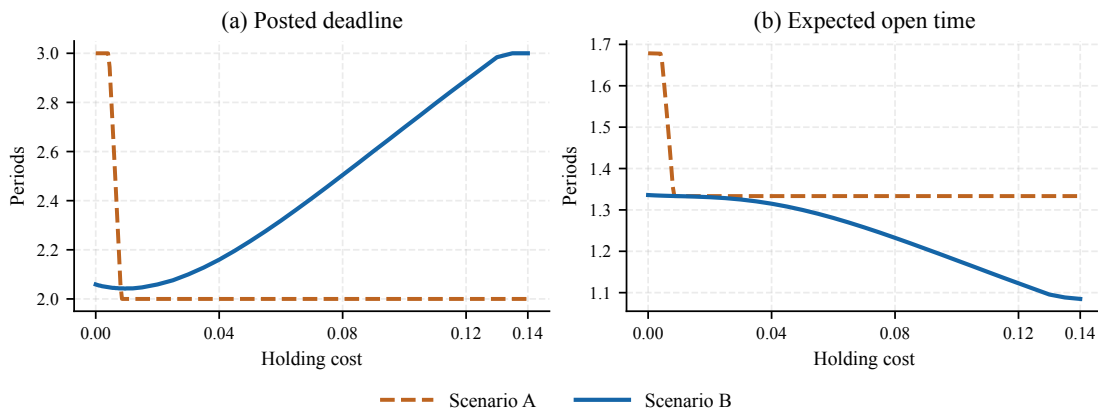


Figure 5 How holding costs affect optimal offers in Scenarios A and B. Left: Optimal deadlines. Right: Expected open times.

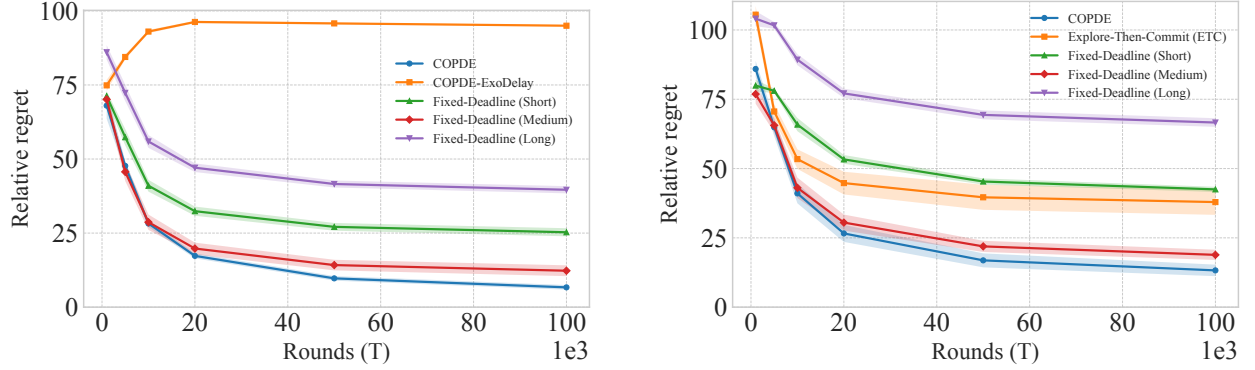


Figure 6 Learning performance and response timing. Left: Exponential environment, comparing COPDE with COPDE-ExoDelay, which learns response timing from customer context only, and fixed-deadline policies. Right: Weibull response times with shape 2. Policies continue to fit an exponential timing model. Each panel reports relative regret against an oracle that knows the actual response-time distribution.

The role of action-dependent response timing. We next ask whether the seller needs to learn how price and deadline affect customer response times. We compare COPDE with COPDE-ExoDelay, which learns response timing from customer context only, excluding price and deadline from the timing features. It therefore assumes that neither choice changes how quickly a customer responds. The variant retains COPDE’s procedures for learning valuations, processing feedback, and eliminating uncompetitive price candidates. It continues to choose both prices and deadlines. A longer deadline still gives customers more time to respond and affects holding costs but it simply does not change their predicted response speed.

Figure 6 (Left) shows substantially higher relative regret for COPDE-ExoDelay. Its timing model cannot capture how a change in price or deadline speeds up or slows down customer responses, and thereby affects both purchases before expiration and holding costs. The comparison shows that ignoring this relationship can be costly even when the seller continues to learn demand and adjust both prices and deadlines.

Sensitivity to the timing specification. We finally test what happens when customer response times do not follow the distribution assumed by the learner. We generate response times from a Weibull distribution, while all policies continue to fit an exponential timing model. The oracle knows the actual Weibull distribution. Figure 6 (Right) shows that COPDE’s relative regret declines as observations accumulate despite the distributional mismatch. By the end of the horizon, COPDE achieves the lowest relative regret among the compared policies. Its final relative regret is nevertheless higher than in the correctly specified exponential environment.

8. Managerial Implications

Our results have implications for how sellers design offers, learn how customers respond to prices and deadlines, and collect the feedback needed to support these decisions.

Shorter deadlines can raise conversion and lower holding costs. A shorter deadline limits the time available to purchase, while potentially accelerating the customer's response. Which effect dominates determines the sales consequence. Appendix EC.7 makes this comparison explicit. In this exponential urgency family, a shorter deadline can increase conversion when it accelerates customer response enough to outweigh the shorter response window, even with price and intrinsic valuation held fixed. Expected open time also falls. Thus a shorter deadline can increase sales revenue and reduce holding costs simultaneously. In Figure 6 (Left), COPDE-ExoDELAY incurs substantially higher regret despite adjusting both prices and deadlines, illustrating the cost of omitting their effects on response timing.

Higher holding costs need not imply shorter deadlines. The announced deadline caps the duration of an offer, but its actual duration also depends on how quickly the customer responds. This distinction matters when the cost of waiting changes. Fix a customer context and hold the demand and timing primitives and the feasible actions constant. Let $a_i = (p_i, \mu_i)$ be an optimal offer at holding-cost rate $\bar{h}_i$, with $\bar{h}_2 > \bar{h}_1$, and write $R(a)$ and $C(a)$ for expected sales revenue and expected open time. Optimality gives

$$R(a_1) - \bar{h}_1 C(a_1) \geq R(a_2) - \bar{h}_1 C(a_2),$$

$$R(a_2) - \bar{h}_2 C(a_2) \geq R(a_1) - \bar{h}_2 C(a_1).$$

Adding these inequalities yields $C(a_2) \leq C(a_1)$. Hence optimal expected open time cannot increase as the holding-cost rate rises, even when price and deadline are reoptimized jointly. The model imposes no corresponding monotonicity on the posted deadline, since both price and deadline can change response speed. Operational comparisons across holding-cost regimes should therefore track expected open time alongside the announced deadline. This comparison concerns optimal offers when demand and response timing are known, whereas an online policy must also account for the cost of learning them.

Record how offers resolve, including those that end without a purchase. Offers that generate no sale can still occupy inventory or capacity until they are resolved or expire. Two policies with similar conversion can therefore have different economic performance. Proposition 1 gives a sharper statement: even at a fixed price, two environments can have identical purchase-before-expiry probabilities at every available deadline and yet have different optimal deadlines because their

expected open times differ. Purchase counts alone cannot reveal this difference. The relevant performance measure is sales revenue net of the cost of keeping offers active, including offers that end without a purchase. Within our observation model, a resolved nonpurchase ends this exposure and supplies a deadline-meeting observation, whereas an unresolved offer remains open until expiration. Retaining the distinction between resolution and expiration therefore supplies the timing information used by COPDE to evaluate costs and correct demand estimates for censoring.

9. Conclusion

Price and deadline choices jointly shape an offer's profitability and the information available for subsequent learning. Our framework captures this connection by accounting for how the announced deadline affects response timing and the observation of buy–reject outcomes.

We develop COPDE to learn prices and deadlines under this information structure. The algorithm combines timing estimation with censoring-adjusted valuation learning and pools demand information across contexts through price offsets. Under the stated assumptions, it achieves regret of $\tilde{O}(T^{2/3})$ while learning from delayed and deadline-censored feedback. The deadline-augmented extension allows the announced deadline to shift the latent buy–reject margin directly, complementing the timing channel in the baseline model.

These results connect deadline design to the measurement of customer response and the economic cost of waiting. Evaluating an offer requires accounting for the time consumed by both successful and unsuccessful offers, as well as how customers respond to the announced deadline. Further work could examine these decisions with shared inventory or service capacity, where accepting one offer changes the opportunities available to other customers. A second direction is to study how sellers can detect changes in customer demand and response timing and adapt their prices and deadlines accordingly.

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Appendix

EC.1. Additional Algorithmic Details

Algorithm 1 in the main text gives the main COPDE loop. This appendix expands the two subroutines used there: `PROCESSEVENTS`, which processes delayed feedback when it matures, and `ELIMINATE`, which recomputes the current non-eliminated offset set and deadline best responses. The bookkeeping convention is that offset-level statistics (N_k, M_k, E_k) are maintained across all contexts, while the non-eliminated offset set $\mathcal{K}_t$ is recomputed at the current context.

EC.1.1. `PROCESSEVENTS`: event-driven updates

The routine `PROCESSEVENTS` processes all previously unprocessed feedback that has matured before round t . After processing, the unindexed parameter estimates and design matrices in the pseudocode are those indexed by $t - 1$ in the analysis. Mode 2 events update the timing estimator, Mode 1 events update the valuation estimator through the stored IPW denominator, and Mode 0 events update the offset-level demand and exposure statistics.

When an offer is posted, COPDE stores the quantities needed for later event processing. In Mode 1, it stores

$$q_s = \widehat{\Psi}_{s-1}^c(x_s, p_s, \bar{\mu}),$$

and in Mode 0, it stores

$$e_s = \widehat{\Psi}_{s-1}^c(x_s, p_s, \mu_s),$$

where

$$\widehat{\Psi}_{s-1}^c(x, p, \mu) := \text{clip}\left(\widehat{\Psi}_{s-1}(x, p, \mu), \pi_{\min}/2, 1\right).$$

These stored values are not recomputed when the corresponding feedback matures.

The pseudocode of `PROCESSEVENTS` is given in Algorithm 2.

EC.1.2. `ELIMINATE`: offset elimination with deadline best response

The routine `ELIMINATE` computes the feasible offsets at the current context, constructs confidence bounds for each feasible offset–deadline pair, and returns the current non-eliminated offset set. The set $\mathcal{K}_t$ is context-specific; learning about the latent demand curve is pooled through the offset-level statistics (N_k, M_k, E_k) .

The pseudocode of `ELIMINATE` is given in Algorithm 3.

Algorithm 2: $\text{PROCESSEVENTS}(t)$: event-driven updates before round t

Input: current time t and stored logs for past offers, including $i_s, x_s, p_s, \mu_s, k_s, \ell_s, q_s, e_s$.

Output: updated $(V, b, \hat{\theta}, V_h, \hat{\psi}, M, E)$.

foreach *unprocessed past offer* s with $s + \mu_s < t$ **do**

Observe $w_s = \mathbb{I}\{d_s \leq \mu_s\}$.

Set

$$B_s \leftarrow \begin{cases} y_s, & w_s = 1, \\ 0, & w_s = 0. \end{cases}$$

if $i_s = 1$ **then**

Retrieve the stored IPW denominator q_s .

Set

$$\tilde{y}_s^{\text{ipw}} \leftarrow \underline{p} + (\bar{p} - \underline{p}) \frac{B_s}{q_s}, \quad z_s \leftarrow x_s \tilde{y}_s^{\text{ipw}}.$$

Update

$$V \leftarrow V + x_s x_s^\top, \quad b \leftarrow b + z_s, \quad \hat{\theta} \leftarrow V^{-1} b.$$

else if $i_s = 2$ **then**

Add $(w_s, h(x_s, p_s, \mu_s), \mu_s)$ to the Mode 2 timing sample.

Recompute $\hat{\psi}$ by the regularized timing likelihood and set

$$V_h \leftarrow \lambda I_{d'} + \underline{\kappa}_\ell \sum_{r: i_r=2, r+\mu_r < t} h_r h_r^\top.$$

else if $i_s = 0$ **then**

Retrieve the offset k_s and stored exposure e_s .

Update

$$M_{k_s} \leftarrow M_{k_s} + B_s, \quad E_{k_s} \leftarrow E_{k_s} + e_s.$$

end

Mark offer s as processed.

end

return updated $(V, b, \hat{\theta}, V_h, \hat{\psi}, M, E)$.

Algorithm 3: ELIMINATE($x_t, \hat{\theta}, \hat{\psi}, \{N, M, E\}$)**Input:** context x_t , estimates $\hat{\theta}, \hat{\psi}$, offset counters and exposures.**Output:** current non-eliminated offset set $\mathcal{K}_t$ and pairwise UCB values.

Form the feasible offset set

$$\mathcal{A}_t \leftarrow \left\{ k \in [K] : x_t^\top \hat{\theta} + \delta_k \in [\underline{p}, \overline{p}] \right\}.$$

foreach $k \in \mathcal{A}_t$ **do**

Set

$$p_{t,k} \leftarrow x_t^\top \hat{\theta} + \delta_k.$$

Form the exposure-normalized demand estimate

$$\widehat{D}_k \leftarrow \left\lfloor \frac{M_k}{\max\{1, E_k\}} \right\rfloor_{[0,1]}.$$

foreach $\ell \in [L]$ **do**

Compute

$$\widehat{r}_t(k, \ell) \leftarrow p_{t,k} \widehat{D}_k \widehat{\Psi}_{t-1}(x_t, p_{t,k}, \mu_\ell) - \bar{h} \widehat{C}_{t-1}(x_t, p_{t,k}, \mu_\ell).$$

Compute the confidence radius $b_t(k, \ell)$ as in (16).

Set

$$\text{UCB}_t(k, \ell) \leftarrow \widehat{r}_t(k, \ell) + b_t(k, \ell), \quad \text{LCB}_t(k, \ell) \leftarrow \widehat{r}_t(k, \ell) - b_t(k, \ell).$$

end

Set

$$\text{UCB}_t(k) \leftarrow \max_{\ell \in [L]} \text{UCB}_t(k, \ell), \quad \text{LCB}_t(k) \leftarrow \max_{\ell \in [L]} \text{LCB}_t(k, \ell).$$

end

Set

$$\mathcal{K}_t \leftarrow \left\{ k \in \mathcal{A}_t : \text{UCB}_t(k) \geq \max_{j \in \mathcal{A}_t} \text{LCB}_t(j) \right\}.$$

return $\mathcal{K}_t$ and $\{\text{UCB}_t(k, \ell)\}_{k, \ell}$.

EC.2. Experimental Details

EC.2.1. Learning environment and oracle

The learning experiments use five customer characteristics, $x \sim \mathcal{N}(0, I_5)$. Valuation is $u = x^\top \theta + \xi$, where the independent Gaussian shock ξ has mean zero and standard deviation 0.01. For price p and deadline μ , the response-rate parameter is $\rho(x, p, \mu) = \exp((x^\top, p, \mu)\psi)$. Contexts are not normalized and valuations are not truncated. We use

$$\theta = (-0.517246, -0.192328, 0.673501, 0.101436, 0.481221)^\top$$

and

$$\psi = (0.305268, -0.336668, 0.286674, -0.167468, -0.170533, 0.051398, -0.807165)^\top.$$

All policies choose prices in $[0, 1]$. Short, Medium, and Long use deadlines 0.5, 1.5, and 2.5, respectively, in every learning mode.

The learning simulations use continuous response times. Let $\Psi(x, p, \mu)$ be the probability that the customer responds before the deadline and $C(x, p, \mu)$ the expected time until the customer responds or the offer expires. In the exponential environment, response time is $\tau \sim \text{Exp}(\rho)$, giving

$$\Psi(x, p, \mu) = 1 - e^{-\rho\mu}, \quad C(x, p, \mu) = \frac{1 - e^{-\rho\mu}}{\rho}.$$

In the Weibull experiment, $\tau = E^{1/2}/\rho$ with $E \sim \text{Exp}(1)$, so

$$\Psi(x, p, \mu) = 1 - e^{-(\rho\mu)^2}, \quad C(x, p, \mu) = \frac{\sqrt{\pi}}{2\rho} \text{erf}(\rho\mu).$$

Here erf denotes the error function. In both cases, expected profit is $p\Phi((x^\top \theta - p)/0.01)\Psi(x, p, \mu) - \bar{h}C(x, p, \mu)$, where Φ is the standard normal distribution function and $\bar{h}$ is the holding cost per unit of time. The first term is expected sales revenue and the second is the expected holding cost. All learning methods fit an exponential timing model, including in the Weibull experiment.

Offers posted at round t supply feedback at $t + \lceil \mu \rceil$, even if the customer responds earlier. Learners observe whether the customer responded before the deadline and, if so, whether the customer purchased. Learners receive neither the true parameters and noise scale nor the oracle's decisions.

EC.2.2. Comparison policies

Fixed-Deadline policies continue learning prices for each customer context while keeping their prescribed deadline. During its first 2,500 offers, ETC draws prices uniformly from $[0, 1]$ and

deadlines uniformly from the shared menu. After all these offers reach their deadlines, it estimates response timing, fits a ridge-regression valuation score from customer context, and estimates demand with a kernel method. It then holds these estimates fixed and chooses the price and deadline with the highest estimated profit for each new context. ETC therefore commits to its estimated models while continuing to tailor offers to customers. Because ETC and COPDE use different estimators, their comparison also reflects these estimation differences.

EC.2.3. Replication and aggregation

The baseline, Weibull, and ExoDelay comparisons use seeds 41000–41029, with 100,000 arrivals per seed and method. Within each seed, all methods receive the same sequences of customer contexts, valuation shocks, and timing shocks and these three sequences are generated independently. The ExoDelay panel uses the baseline COPDE and Fixed-Deadline results and the same exponential oracle.

For each checkpoint n , relative regret is

$$100 \frac{\sum_{t=1}^n \{r^*(x_t) - r(x_t; p_t, \mu_t)\}}{\sum_{t=1}^n r^*(x_t)},$$

Here $r(x_t; p_t, \mu_t)$ is the expected profit of the policy's offer and $r^*(x_t)$ is the oracle reference, both evaluated under the actual valuation and response-time distributions. We compute relative regret separately for each run, then report its mean and 1.96*standard error across runs. Checkpoints are 1,000, 5,000, 10,000, 20,000, 50,000, and 100,000.

EC.2.4. Holding-cost response scenarios

The experiment uses $x = (1, z)$, with $z \sim \text{Unif}[-1, 1]$, and valuation $u = 0.5 + 0.02z + \xi$, where $\xi \sim \text{Unif}[-0.4, 0.4]$ is independent of the timing shock. Prices lie in $[0, 1]$ and deadlines in $\{1, 2, 3\}$. Here response time is integer valued: $\tau = \lceil \tilde{\tau} \rceil$, with $\tilde{\tau} \sim \text{Exp}(\rho_\mu(z, p))$, where

$$\rho_1(z, p) = \rho_2(z, p) = (\log 3)e^{0.1z}, \quad \rho_3(z, p) = 0.85e^{0.1z + \kappa(p_{\text{ref}} - p)}$$

and $p_{\text{ref}} = 0.24 + \log((\log 20)/0.85)/15 \approx 0.323981$. Scenario A uses $\kappa = 1$; Scenario B uses $\kappa = 15$. Their response rates coincide at the reference price. Price affects timing at deadline 3 in both scenarios, and does not affect timing at deadlines 1 and 2.

The response probability and expected open duration are

$$\Psi(x, p, \mu) = 1 - e^{-\mu \rho_\mu(z, p)}, \quad C(x, p, \mu) = \sum_{j=0}^{\mu-1} e^{-j \rho_\mu(z, p)}.$$

At each cost, we jointly maximize $p \Pr(u \geq p \mid z) \Psi(x, p, \mu) - \bar{h}C(x, p, \mu)$ over price and deadline, keeping each scenario's valuation and response-time models fixed. We average optimal decisions and their expected open times over 1,001 equally spaced midpoint values of z in $[-1, 1]$. For each context, numerical lower and upper bounds on optimal expected profit differ by at most 10^{-6} . The scenarios use the same 51 cost values in $[0, 0.14]$, with denser spacing near the initial deadline change. The seller knows all model parameters in this experiment and optimizes offers directly.

EC.3. Verification of Assumption 1 for Exponential Timing

Let

$$a := \mu e^u.$$

For $w = 0$, the Bernoulli timing loss is

$$\ell_\mu(0, u) = -\log(1 - g_\mu(u)) = a,$$

so

$$\partial_u \ell_\mu(0, u) = a, \quad \partial_{uu}^2 \ell_\mu(0, u) = a.$$

For $w = 1$,

$$\ell_\mu(1, u) = -\log g_\mu(u) = -\log(1 - e^{-a}),$$

and direct differentiation gives

$$\begin{aligned} \partial_u \ell_\mu(1, u) &= -\frac{a}{e^a - 1}, \\ \partial_{uu}^2 \ell_\mu(1, u) &= \frac{a\{(a-1)e^a + 1\}}{(e^a - 1)^2}. \end{aligned}$$

Since $(a-1)e^a + 1 > 0$ for every $a > 0$, both curvature expressions are strictly positive. Moreover, because

$$a = \mu e^u \in [\underline{\mu} e^{-Bz}, \bar{\mu} e^{Bz}],$$

the score and curvature terms are continuous on a compact subset of $(0, \infty)$. Hence the score is uniformly bounded and the loss curvature is uniformly bounded below by a positive constant. Therefore, the exponential timing specification satisfies the smoothness and timing-information conditions in Assumption 1.

EC.4. Proofs for the Regret Upper Bound (Section 5)

This appendix contains the proofs of the technical results used in the regret analysis. We use the notation introduced in the main text. Feedback from an offer posted at time s is treated as available before round t only if $s + \mu_s < t$.

EC.4.1. Exploration-count bound

We first bound the number of rounds spent in the two dedicated exploration modes. Recall that COPDE checks timing uncertainty before valuation uncertainty. Thus define

$$G_2 := \left\{ t \leq T : \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_{h,t-1}^{-1}} > \eta_h \right\},$$

and

$$G_1 := \left\{ t \leq T : t \notin G_2, \|x_t\|_{V_{t-1}^{-1}} > \eta \right\}.$$

Here G_2 is the set of timing-exploration rounds and G_1 is the set of valuation-exploration rounds.

LEMMA EC.1. *Suppose feedback delays are bounded by $\bar{\mu}$, and let*

$$V_t = \lambda I_d + \sum_{s \leq t: i_s=1, s+\bar{\mu} \leq t} x_s x_s^\top$$

be the valuation design matrix based on feedback matured by calendar time t . Let

$$V_{h,t} = \lambda I_{d'} + \underline{\kappa}_\ell \sum_{s \leq t: i_s=2, s+\mu_s \leq t} h_s h_s^\top$$

be the corresponding timing design matrix. The round- t decision uses V_{t-1} and $V_{h,t-1}$. Then

$$|G_1| \leq \frac{2(1+\bar{\mu})d}{\eta^2} \log \left(1 + \frac{TB^2}{\lambda d} \right),$$

and, whenever $\underline{\kappa}_\ell \eta_h^2 \leq 1$,

$$|G_2| \leq \frac{2(1+\bar{\mu})d'}{\underline{\kappa}_\ell \eta_h^2} \log \left(1 + \frac{T \underline{\kappa}_\ell B_h^2}{\lambda d'} \right).$$

Consequently,

$$|G_1 \cup G_2| \leq \frac{2(1+\bar{\mu})d}{\eta^2} \log \left(1 + \frac{TB^2}{\lambda d} \right) + \frac{2(1+\bar{\mu})d'}{\underline{\kappa}_\ell \eta_h^2} \log \left(1 + \frac{T \underline{\kappa}_\ell B_h^2}{\lambda d'} \right).$$

Proof. We prove the two bounds separately.

First consider G_1 . Let

$$q := \bar{\mu} + 1.$$

Partition G_1 into residue classes

$$G_{1,r} := \{t \in G_1 : t \equiv r \pmod{q}\}, \quad r = 0, \dots, q-1.$$

Fix one residue class r , and enumerate its elements as

$$t_1 < t_2 < \cdots < t_m.$$

Since all times in the same residue class are separated by at least $\bar{\mu} + 1$, every earlier valuation-exploration observation in this residue class satisfies $s + \bar{\mu} \leq t_j - 1$ and has matured before the next one is selected. Hence, before the decision at time t_j , the valuation design matrix contains all previous design vectors $x_{t_1}, \dots, x_{t_{j-1}}$ from this residue class.

Define

$$M_0 := \lambda I_d, \quad M_j := M_{j-1} + x_{t_j} x_{t_j}^\top.$$

Since $V_{t_{j-1}} \succeq M_{j-1}$, we have

$$\|x_{t_j}\|_{V_{t_{j-1}}^{-1}} \leq \|x_{t_j}\|_{M_{j-1}^{-1}}.$$

For $t_j \in G_1$,

$$\|x_{t_j}\|_{V_{t_{j-1}}^{-1}} > \eta,$$

and therefore

$$\|x_{t_j}\|_{M_{j-1}^{-1}} > \eta.$$

Assuming $\eta^2 \leq 1$, the elliptical-potential lemma gives

$$m\eta^2 \leq \sum_{j=1}^m \min\{1, \|x_{t_j}\|_{M_{j-1}^{-1}}^2\} \leq 2 \log \frac{\det(M_m)}{\det(M_0)}.$$

Using $\|x_t\| \leq B$ and the standard determinant bound,

$$\log \frac{\det(M_m)}{\det(M_0)} \leq d \log \left(1 + \frac{mB^2}{\lambda d} \right) \leq d \log \left(1 + \frac{TB^2}{\lambda d} \right).$$

Thus,

$$|G_{1,r}| \leq \frac{2d}{\eta^2} \log \left(1 + \frac{TB^2}{\lambda d} \right).$$

Summing over the $q = 1 + \bar{\mu}$ residue classes proves the stated bound for $|G_1|$.

The argument for G_2 is analogous. Again partition G_2 into residue classes modulo $q = \bar{\mu} + 1$. Fix a residue class and enumerate its elements as $t_1 < \cdots < t_m$. At each timing-exploration round t_j , the algorithm chooses

$$(p_{t_j}, \mu_{t_j}) \in \arg \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_{t_j}, p, \mu)\|_{V_{h, t_{j-1}}^{-1}},$$

so the realized feature

$$h_{t_j} := h(x_{t_j}, p_{t_j}, \mu_{t_j})$$

satisfies

$$\|h_{t_j}\|_{V_{h,t_j-1}^{-1}} > \eta_h.$$

Define

$$M_0^h := \lambda I_{d'}, \quad M_j^h := M_{j-1}^h + \underline{\kappa}_\ell h_{t_j} h_{t_j}^\top.$$

As before, by the residue-class construction, earlier observations in the same class satisfy $s + \mu_s \leq t_j - 1$ and have matured before the next decision, so

$$V_{h,t_j-1} \succeq M_{j-1}^h.$$

Therefore

$$\|h_{t_j}\|_{(M_{j-1}^h)^{-1}} > \eta_h,$$

or equivalently

$$\|\sqrt{\underline{\kappa}_\ell} h_{t_j}\|_{(M_{j-1}^h)^{-1}}^2 > \underline{\kappa}_\ell \eta_h^2.$$

If $\underline{\kappa}_\ell \eta_h^2 \leq 1$, the elliptical-potential lemma yields

$$m \underline{\kappa}_\ell \eta_h^2 \leq \sum_{j=1}^m \min \left\{ 1, \|\sqrt{\underline{\kappa}_\ell} h_{t_j}\|_{(M_{j-1}^h)^{-1}}^2 \right\} \leq 2 \log \frac{\det(M_m^h)}{\det(M_0^h)}.$$

Since $\|h_t\| \leq B_h$,

$$\log \frac{\det(M_m^h)}{\det(M_0^h)} \leq d' \log \left(1 + \frac{m \underline{\kappa}_\ell B_h^2}{\lambda d'} \right) \leq d' \log \left(1 + \frac{T \underline{\kappa}_\ell B_h^2}{\lambda d'} \right).$$

Thus,

$$|G_{2,r}| \leq \frac{2d'}{\underline{\kappa}_\ell \eta_h^2} \log \left(1 + \frac{T \underline{\kappa}_\ell B_h^2}{\lambda d'} \right).$$

Summing over the $1 + \bar{\mu}$ residue classes proves the bound for G_2 . □

EC.4.2. Concentration Results

Proof of Lemma 1. Enumerate all Mode 1 probes in posting order as $s_1 < s_2 < \dots$, stopping at the horizon. Since every such probe uses the same deadline $\bar{\mu}$, the probes matured by each calendar time form a prefix of this fixed list:

$$\mathcal{I}_t^\theta = \{s_1, \dots, s_{N_t^\theta}\}.$$

For each posted probe j , define

$$X_j := x_{s_j}, \quad P_j := p_{s_j}, \quad Q_j := q_{s_j} = \widehat{\Psi}_{s_j-1}^c(x_{s_j}, p_{s_j}, \bar{\mu}),$$

and

$$\widetilde{Y}_j := \widetilde{y}_{s_j}^{\text{ipw}} = \underline{p} + R_p \frac{B_{s_j}}{Q_j}.$$

Let

$$\Psi_j := \Psi(X_j, P_j, \bar{\mu}).$$

Since $Q_j \geq \pi_{\min}/2$ by clipping, the pseudo-label $\widetilde{Y}_j$ is uniformly bounded:

$$\widetilde{Y}_j \in \left[\underline{p}, \underline{p} + \frac{2R_p}{\pi_{\min}} \right].$$

We first derive the moment decomposition. For the analysis, let $\mathcal{G}_{j-1}$ contain the history just before the randomized price P_j is drawn, including X_j , the decision to enter Mode 1, and all customer shocks from earlier offers, whether or not their feedback has matured. This enlarged history is used only in the proof. By the stochastic-primitives assumption and fresh policy randomization, conditional on $\mathcal{G}_{j-1}$ the price P_j is uniform on $[\underline{p}, \bar{p}]$, and the current customer shocks retain their stated conditional laws.

Define the ideal, oracle-weighted pseudo-label

$$\widetilde{Y}_j^\star := \underline{p} + R_p \frac{B_{s_j}}{\Psi_j}.$$

Using the conditional separation of the purchase and timing components,

$$\mathbb{E}[B_{s_j} \mid \mathcal{G}_{j-1}, P_j] = D(P_j - X_j^\top \theta) \Psi_j.$$

Therefore,

$$\mathbb{E}[\widetilde{Y}_j^\star \mid \mathcal{G}_{j-1}, P_j] = \underline{p} + R_p D(P_j - X_j^\top \theta).$$

Taking expectation over the uniform random price P_j gives

$$\mathbb{E}[\tilde{Y}_j^\star \mid \mathcal{G}_{j-1}] = \underline{p} + R_p \mathbb{E}_{P_j} \left[D(P_j - X_j^\top \theta) \mid \mathcal{G}_{j-1} \right].$$

Since $P_j \sim \text{Unif}[\underline{p}, \bar{p}]$, and the valuation $u_j = X_j^\top \theta + \xi_{s_j}$ lies in $[\underline{p}, \bar{p}]$ almost surely on the operating domain,

$$\mathbb{E}_{P_j} [\mathbb{I}\{u_j \geq P_j\} \mid u_j] = \frac{u_j - \underline{p}}{R_p}.$$

Taking expectation over ξ_{s_j} and using $\mathbb{E}[\xi_{s_j}] = 0$, we obtain

$$\mathbb{E}_{P_j} \left[D(P_j - X_j^\top \theta) \mid \mathcal{G}_{j-1} \right] = \frac{X_j^\top \theta - \underline{p}}{R_p}.$$

Hence

$$\mathbb{E}[\tilde{Y}_j^\star \mid \mathcal{G}_{j-1}] = X_j^\top \theta.$$

For each possible probe price, write

$$Q_j(p) := \widehat{\Psi}_{s_{j-1}}^c(X_j, p, \bar{\mu}), \quad \Psi_j(p) := \Psi(X_j, p, \bar{\mu}),$$

so that $Q_j = Q_j(P_j)$ and $\Psi_j = \Psi_j(P_j)$. Both functions are fixed given $\mathcal{G}_{j-1}$. On $\mathcal{E}_\psi^\theta(\varepsilon_\psi)$, the pre-price history of each probe satisfies

$$\sup_{p \in [\underline{p}, \bar{p}]} |Q_j(p) - \Psi_j(p)| \leq \varepsilon_\psi, \quad Q_j(p) \geq \pi_{\min}/2.$$

Hence, at every such history,

$$\sup_{p \in [\underline{p}, \bar{p}]} \left| \frac{\Psi_j(p)}{Q_j(p)} - 1 \right| \leq \frac{2\varepsilon_\psi}{\pi_{\min}}.$$

The bound therefore holds throughout the price integral in the conditional expectation below, without conditioning that expectation on the future timing-good event:

$$\begin{aligned} \left| \mathbb{E}[\tilde{Y}_j - \tilde{Y}_j^\star \mid \mathcal{G}_{j-1}] \right| &= R_p \left| \mathbb{E} \left[B_{s_j} \left(\frac{1}{Q_j} - \frac{1}{\Psi_j} \right) \mid \mathcal{G}_{j-1} \right] \right| \\ &= R_p \left| \mathbb{E} \left[D(P_j - X_j^\top \theta) \left(\frac{\Psi_j}{Q_j} - 1 \right) \mid \mathcal{G}_{j-1} \right] \right| \\ &\leq \frac{2R_p}{\pi_{\min}} \varepsilon_\psi. \end{aligned}$$

Define

$$b_j^\theta := \mathbb{E}[\tilde{Y}_j \mid \mathcal{G}_{j-1}] - X_j^\top \theta$$

and

$$v_j := \tilde{Y}_j - \mathbb{E}[\tilde{Y}_j \mid \mathcal{G}_{j-1}].$$

Then

$$\tilde{Y}_j = X_j^\top \theta + b_j^\theta + v_j,$$

with

$$\mathbb{E}[v_j \mid \mathcal{G}_{j-1}] = 0, \quad |b_j^\theta| \leq C_{\text{ipw}} \varepsilon_\psi, \quad C_{\text{ipw}} = \frac{2R_p}{\pi_{\min}}.$$

Next, since $\tilde{Y}_j$ lies in an interval of length at most $2R_p/\pi_{\min}$, Hoeffding's lemma implies that v_j is conditionally R_θ^{ipw} -sub-Gaussian with

$$R_\theta^{\text{ipw}} = \frac{R_p}{\pi_{\min}}.$$

Define

$$\tilde{V}_n := \lambda I_d + \sum_{j=1}^n X_j X_j^\top, \quad \tilde{S}_n := \sum_{j=1}^n X_j v_j.$$

Applying the self-normalized martingale concentration inequality of Abbasi-Yadkori et al. (2011) to the adaptively stopped sequence (X_j, v_j) gives an event of probability at least $1 - \zeta_\theta$, unconditionally, on which the following holds for every probe prefix $n \leq N_T^\theta$:

$$\|\tilde{S}_n\|_{\tilde{V}_n^{-1}} \leq R_\theta^{\text{ipw}} \sqrt{2 \ln \left(\frac{\det(\tilde{V}_n)^{1/2} \det(\lambda I_d)^{-1/2}}{\zeta_\theta} \right)}.$$

Using

$$\det(\tilde{V}_n) = \det(\lambda I_d) \det \left(I_d + \lambda^{-1} \sum_{j=1}^n X_j X_j^\top \right),$$

and

$$\det(I + A) \leq \left(1 + \frac{\text{tr}(A)}{d} \right)^d \quad (A \succeq 0),$$

together with $\|X_j\| \leq B$, we obtain

$$\|\tilde{S}_n\|_{\tilde{V}_n^{-1}} \leq R_\theta^{\text{ipw}} \sqrt{2 \left[d \ln \left(1 + \frac{nB^2}{\lambda d} \right) + \ln \frac{1}{\zeta_\theta} \right]} \quad \forall n \leq N_T^\theta.$$

Now define

$$\tilde{b}_n := \sum_{j=1}^n X_j \tilde{Y}_j, \quad \tilde{\theta}_n := \tilde{V}_n^{-1} \tilde{b}_n.$$

Using the decomposition

$$\tilde{Y}_j = X_j^\top \theta + b_j^\theta + \nu_j,$$

we have

$$\tilde{b}_n = (\tilde{V}_n - \lambda I_d) \theta + \tilde{S}_n + \tilde{B}_n, \quad \tilde{B}_n := \sum_{j=1}^n X_j b_j^\theta.$$

Therefore,

$$\tilde{\theta}_n - \theta = \tilde{V}_n^{-1} (\tilde{S}_n + \tilde{B}_n - \lambda \theta).$$

For any $x \in \mathbb{R}^d$,

$$\left| x^\top (\tilde{\theta}_n - \theta) \right| \leq \|x\|_{\tilde{V}_n^{-1}} \left(\|\tilde{S}_n\|_{\tilde{V}_n^{-1}} + \|\tilde{B}_n\|_{\tilde{V}_n^{-1}} + \sqrt{\lambda} \|\theta\| \right).$$

It remains to bound the bias term. Let $X \in \mathbb{R}^{n \times d}$ be the matrix with rows $X_j^\top$, and let $b = (b_1^\theta, \dots, b_n^\theta)^\top$. Then

$$\tilde{B}_n = X^\top b,$$

and

$$\|\tilde{B}_n\|_{\tilde{V}_n^{-1}}^2 = b^\top X \tilde{V}_n^{-1} X^\top b.$$

Since

$$X \tilde{V}_n^{-1} X^\top = X(\lambda I_d + X^\top X)^{-1} X^\top \preceq I_n,$$

we have

$$\|\tilde{B}_n\|_{\tilde{V}_n^{-1}} \leq \|b\|_2 \leq C_{\text{ipw}} \varepsilon_\psi \sqrt{n}.$$

Intersect this noise-concentration event with $\mathcal{E}_\psi^\theta(\varepsilon_\psi)$. Their intersection has probability at least $1 - \zeta_\theta - \zeta_\psi$, and the preceding inequalities give, simultaneously for all $n \leq N_T^\theta$,

$$\left| x^\top (\tilde{\theta}_n - \theta) \right| \leq \|x\|_{\tilde{V}_n^{-1}} \left[R_\theta^{\text{ipw}} \sqrt{2 \left[d \ln \left(1 + \frac{n B^2}{\lambda d} \right) + \ln \frac{1}{\zeta_\theta} \right]} + \sqrt{\lambda} \|\theta\| + C_{\text{ipw}} \varepsilon_\psi \sqrt{n} \right].$$

Finally, at calendar time t ,

$$V_t = \tilde{V}_{N_t^\theta}, \quad \hat{\theta}_t = \tilde{\theta}_{N_t^\theta}.$$

Taking $n = N_t^\theta$ yields

$$\left| x^\top (\hat{\theta}_t - \theta) \right| \leq \left[\beta_t^{\theta, \text{ipw}}(\zeta_\theta) + C_{\text{ipw}} \varepsilon_\psi \sqrt{N_t^\theta} \right] \|x\|_{V_t^{-1}}.$$

The case $t = 0$ holds deterministically: $V_0 = \lambda I_d$, $\hat{\theta}_0 = 0$, and the ridge term bounds the prediction error. This proves the lemma. $\square$

COROLLARY EC.1. *Let $\beta_t^{\theta, \text{ipw}}(\zeta)$ be as in Lemma 1. For any target $\varepsilon_\theta > 0$, define*

$$\eta := \frac{\varepsilon_\theta}{\beta_T^{\theta, \text{ipw}}(\zeta)}.$$

Then, on the event of Lemma 1 and the Mode 1 timing-good event, every round t satisfying $\|x_t\|_{V_{t-1}^{-1}} \leq \eta$ also satisfies

$$\left| x_t^\top (\widehat{\theta}_{t-1} - \theta) \right| \leq \varepsilon_\theta + C_{\text{ipw}} \varepsilon_\psi \sqrt{N_{t-1}^\theta} \eta,$$

which consequently guarantees

$$\left| x_t^\top (\widehat{\theta}_{t-1} - \theta) \right| \leq \varepsilon_\theta + C_{\text{ipw}} \varepsilon_\psi \sqrt{2(1 + \bar{\mu})d \log \left(1 + \frac{TB^2}{\lambda d} \right)}.$$

Proof. On the event of Lemma 1,

$$\left| x_t^\top (\widehat{\theta}_{t-1} - \theta) \right| \leq \left[\beta_{t-1}^{\theta, \text{ipw}}(\zeta) + C_{\text{ipw}} \varepsilon_\psi \sqrt{N_{t-1}^\theta} \right] \|x_t\|_{V_{t-1}^{-1}}.$$

Since $\beta_{t-1}^{\theta, \text{ipw}}(\zeta) \leq \beta_T^{\theta, \text{ipw}}(\zeta)$ and $\|x_t\|_{V_{t-1}^{-1}} \leq \eta$, the first displayed claim follows. The second claim follows by substituting the exploration-count bound $N_{t-1}^\theta \eta^2 \leq 2(1 + \bar{\mu})d \log(1 + TB^2/(\lambda d))$ established in Lemma EC.1. $\square$

EC.4.2.1. Proof of Lemma 2 *Proof of Lemma 2.* Fix a deterministic calendar time $t \in \{1, \dots, T\}$. For each offer $s \leq t$, define

$$J_s^{(t)} := \mathbb{I}\{i_s = 2, s + \mu_s \leq t\}, \quad \dot{\ell}_s(\psi) := \partial_z \ell_{\mu_s}(w_s, h_s^\top \psi).$$

For the analysis, let $\mathcal{G}_s^\psi$ contain the history just after the context, mode, price, and deadline of offer s have been chosen, enlarged by all customer shocks from offers $r < s$. The current customer's shocks are excluded. This action-time history is used only in the proof. The stochastic-primitives assumption gives

$$\mathbb{E}[w_s \mid \mathcal{G}_s^\psi] = g_{\mu_s}(h_s^\top \psi).$$

Since

$$\partial_z \ell_\mu(w, z) = \frac{g'_\mu(z)}{g_\mu(z)(1 - g_\mu(z))} (g_\mu(z) - w),$$

the score satisfies

$$\mathbb{E}[\dot{\ell}_s(\psi) \mid \mathcal{G}_s^\psi] = 0, \quad |\dot{\ell}_s(\psi)| \leq R_\ell.$$

For fixed t , both $J_s^{(t)}$ and h_s are known before the current shocks. Thus the masked score increments are martingale differences in posting order, even when feedback matures in a different order.

Define

$$S_t := \sum_{s \leq t} J_s^{(t)} \dot{\ell}_s(\psi) h_s, \quad V_{h,t} = \lambda I_{d'} + \kappa_\ell \sum_{s \leq t} J_s^{(t)} h_s h_s^\top.$$

Apply the self-normalized martingale inequality to the predictable vectors $\sqrt{\kappa_\ell} J_s^{(t)} h_s$ and the conditionally centered noises $\dot{\ell}_s(\psi)/\sqrt{\kappa_\ell}$. The latter are conditionally $R_\ell/\sqrt{\kappa_\ell}$ -sub-Gaussian by the bounded-score condition. With failure probability at most ζ/T ,

$$\|S_t\|_{V_{h,t}^{-1}} \leq \frac{R_\ell}{\sqrt{\kappa_\ell}} \sqrt{2 \ln \left(\frac{T}{\zeta} \frac{\det(V_{h,t})^{1/2}}{\det(\lambda I_{d'})^{1/2}} \right)}.$$

A union bound over $t = 1, \dots, T$ makes this inequality simultaneous at all positive calendar times with probability at least $1 - \zeta$. No independence between the selected samples at different calendar times is required.

Using $\|h_s\| \leq B_h$ and $\det(I + A) \leq (1 + \text{tr}(A)/d')^{d'}$ for $A \succeq 0$ gives

$$\frac{\det(V_{h,t})}{\det(\lambda I_{d'})} \leq \left(1 + \frac{N_t^\psi \kappa_\ell B_h^2}{\lambda d'} \right)^{d'}.$$

Consequently, on the same event, the following conservative bound holds simultaneously for all $t \leq T$:

$$\|S_t\|_{V_{h,t}^{-1}} \leq \frac{R_\ell}{\sqrt{\kappa_\ell}} \sqrt{2 \left[d' \ln \left(1 + \frac{N_t^\psi \kappa_\ell B_h^2}{\lambda d'} \right) + \ln \frac{T}{\zeta} \right]}.$$

At each calendar time, the regularized objective is

$$Q_t(\vartheta) := \frac{\lambda}{2} \|\vartheta\|_2^2 + \sum_{s \in I_t^\psi} \ell_{\mu_s}(w_s, h_s^\top \vartheta),$$

and $\widehat{\psi}_t$ minimizes Q_t over the convex set Θ_ψ . Since $\psi, \widehat{\psi}_t \in \Theta_\psi$, all scalar indices along the segment joining them lie in $[-B_z, B_z]$. The uniform loss-curvature condition therefore gives

$$\langle \nabla Q_t(\vartheta) - \nabla Q_t(\psi), \vartheta - \psi \rangle \geq \|\vartheta - \psi\|_{V_{h,t}}^2 \quad \forall \vartheta \in \Theta_\psi.$$

The constrained first-order optimality condition implies

$$\langle \nabla Q_t(\widehat{\psi}_t), \psi - \widehat{\psi}_t \rangle \geq 0.$$

Combining these inequalities and using $\nabla Q_t(\psi) = \lambda \psi + S_t$ yields

$$\begin{aligned} \|\widehat{\psi}_t - \psi\|_{V_{h,t}}^2 &\leq -\langle \lambda \psi + S_t, \widehat{\psi}_t - \psi \rangle \\ &\leq \left(\|S_t\|_{V_{h,t}^{-1}} + \sqrt{\lambda} \|\psi\| \right) \|\widehat{\psi}_t - \psi\|_{V_{h,t}}. \end{aligned}$$

Dividing by the final norm when it is nonzero, and treating the zero case trivially, gives

$$\|\widehat{\psi}_t - \psi\|_{V_{h,t}} \leq \|S_t\|_{V_{h,t}^{-1}} + \sqrt{\lambda}\|\psi\| \leq \beta_t^\psi(\zeta).$$

At $t = 0$, the same bound holds deterministically because $V_{h,0} = \lambda I_{d'}$, $\widehat{\psi}_0 = 0$, and the radius includes $\sqrt{\lambda}\|\psi\|$. Thus the bound holds on a single event for all $t = 0, \dots, T$. For every $(x, p, \mu) \in \mathcal{X} \times \mathcal{P} \times \mathcal{M}$, Cauchy–Schwarz now gives

$$\left| h(x, p, \mu)^\top (\widehat{\psi}_t - \psi) \right| \leq \|h(x, p, \mu)\|_{V_{h,t}^{-1}} \beta_t^\psi(\zeta).$$

Finally, the slope bounds on g_μ and c_μ imply

$$\left| \widehat{\Psi}_t(x, p, \mu) - \Psi(x, p, \mu) \right| \leq L_g \beta_t^\psi(\zeta) \|h(x, p, \mu)\|_{V_{h,t}^{-1}},$$

and

$$\left| \widehat{C}_t(x, p, \mu) - C(x, p, \mu) \right| \leq L_C \beta_t^\psi(\zeta) \|h(x, p, \mu)\|_{V_{h,t}^{-1}}.$$

□

COROLLARY EC.2. *Let $\beta_t^\psi(\zeta)$ be as in Lemma 2. For any target $\varepsilon_\psi > 0$, define*

$$\eta_h := \frac{\varepsilon_\psi}{\max\{L_g, L_C\} \beta_T^\psi(\zeta)}.$$

Then, on the event of Lemma 2, any queried triple (x_t, p, μ) satisfying

$$\|h(x_t, p, \mu)\|_{V_{h,t-1}^{-1}} \leq \eta_h$$

also satisfies

$$\left| \widehat{\Psi}_{t-1}(x_t, p, \mu) - \Psi(x_t, p, \mu) \right| \leq \varepsilon_\psi, \quad \left| \widehat{C}_{t-1}(x_t, p, \mu) - C(x_t, p, \mu) \right| \leq \varepsilon_\psi.$$

Proof. By Lemma 2,

$$\left| \widehat{\Psi}_{t-1}(x_t, p, \mu) - \Psi(x_t, p, \mu) \right| \leq L_g \beta_{t-1}^\psi(\zeta) \|h(x_t, p, \mu)\|_{V_{h,t-1}^{-1}},$$

and similarly

$$\left| \widehat{C}_{t-1}(x_t, p, \mu) - C(x_t, p, \mu) \right| \leq L_C \beta_{t-1}^\psi(\zeta) \|h(x_t, p, \mu)\|_{V_{h,t-1}^{-1}}.$$

Since $\beta_{t-1}^\psi(\zeta) \leq \beta_T^\psi(\zeta)$ and the leverage is at most η_h , both bounds are at most ε_ψ . □

EC.4.2.2. Proof of Lemma 3

LEMMA EC.2. *Let*

$$\mathcal{S}_k(t) := \{s < t : i_s = 0, k_s = k, s + \mu_s < t\}$$

be the set of matured Mode 0 offers with offset k available before round t , and define

$$\Lambda_k(t) := \sum_{s \in \mathcal{S}_k(t)} \Psi(x_s, p_s, \mu_s).$$

Let

$$M_k(t) := \sum_{s \in \mathcal{S}_k(t)} B_s, \quad E_k(t) := \sum_{s \in \mathcal{S}_k(t)} e_s,$$

where the stored exposure is

$$e_s := \widehat{\Psi}_{s-1}^c(x_s, p_s, \mu_s), \quad \widehat{\Psi}_{s-1}^c(x, p, \mu) := \text{clip}\left(\widehat{\Psi}_{s-1}(x, p, \mu), \pi_{\min}/2, 1\right).$$

Suppose that, on the Mode 0 confidence event,

$$\left| x_s^\top (\widehat{\theta}_{s-1} - \theta) \right| \leq \varepsilon_{\theta, \text{eff}}, \quad \left| \widehat{\Psi}_{s-1}(x_s, p_s, \mu_s) - \Psi(x_s, p_s, \mu_s) \right| \leq \varepsilon_\psi$$

for all Mode 0 rounds s . Then there exists an event of probability at least $1 - \zeta$ such that, on its intersection with the Mode 0 confidence event, uniformly over all $k \in [K]$ and all $t \leq T$,

$$\left| \widehat{D}_k(t) - D(\delta_k) \right| \leq c_1 \sqrt{\frac{\log(eKT/\zeta)}{\max\{1, E_k(t)\}}} + c_2 \varepsilon_{\theta, \text{eff}} + c_3 \varepsilon_\psi,$$

where

$$\widehat{D}_k(t) := \left[\frac{M_k(t)}{\max\{1, E_k(t)\}} \right]_{[0,1]},$$

and c_1, c_2, c_3 depend only on the operating-domain constants, including $\pi_{\min}^{-1}$.

Proof. Fix an offset k and a time t . For each $s \in \mathcal{S}_k(t)$, write

$$\Psi_s := \Psi(x_s, p_s, \mu_s), \quad e_s = \widehat{\Psi}_{s-1}^c(x_s, p_s, \mu_s),$$

$$D_s := D(p_s - x_s^\top \theta), \quad D_k := D(\delta_k).$$

Since a Mode 0 offer with offset k posts

$$p_s = x_s^\top \widehat{\theta}_{s-1} + \delta_k,$$

we have

$$p_s - x_s^\top \theta = \delta_k + x_s^\top (\widehat{\theta}_{s-1} - \theta).$$

On the valuation-confidence event,

$$\left| x_s^\top (\widehat{\theta}_{s-1} - \theta) \right| \leq \varepsilon_{\theta, \text{eff}}.$$

By Lipschitz continuity of D ,

$$|D_s - D_k| \leq L_D \varepsilon_{\theta, \text{eff}}. \quad (\text{EC.1})$$

Under the factorization in (8),

$$\mathbb{E}[B_s \mid \mathcal{G}_s] = D_s \Psi_s,$$

where $\mathcal{G}_s$ denotes the sigma-field after x_s is observed and the action (i_s, k_s, p_s, μ_s) has been chosen, but before the purchase-before-expiry outcome B_s is realized. Define

$$Z_s := B_s - \mathbb{E}[B_s \mid \mathcal{G}_s].$$

Then Z_s is a bounded martingale difference with respect to the action-time filtration, and

$$|Z_s| \leq 1, \quad \text{Var}(Z_s \mid \mathcal{G}_s) \leq \mathbb{E}[B_s^2 \mid \mathcal{G}_s] = D_s \Psi_s \leq \Psi_s.$$

For fixed k and t , define the predictable indicator

$$I_s^{k,t} := \mathbb{I}\{s < t, i_s = 0, k_s = k, s + \mu_s < t\}.$$

The sequence $I_s^{k,t} Z_s$ is again a bounded martingale-difference sequence, with predictable variance bounded by

$$\sum_{s=1}^T I_s^{k,t} \text{Var}(Z_s \mid \mathcal{G}_s) \leq \sum_{s=1}^T I_s^{k,t} \Psi_s = \Lambda_k(t).$$

Applying Freedman's inequality to $I_s^{k,t} Z_s$, followed by a union bound over $k \in [K]$ and $t \leq T$, gives an event of probability at least $1 - \zeta$ on which, simultaneously for all k, t ,

$$\left| \sum_{s \in \mathcal{S}_k(t)} Z_s \right| \leq C_1 \sqrt{\Lambda_k(t) \log(eKT/\zeta)} + C_2 \log(eKT/\zeta) \quad (\text{EC.2})$$

for universal constants $C_1, C_2 < \infty$.

Next,

$$M_k(t) - D_k \Lambda_k(t) = \sum_{s \in \mathcal{S}_k(t)} Z_s + \sum_{s \in \mathcal{S}_k(t)} (D_s - D_k) \Psi_s.$$

Using (EC.1), we obtain

$$\left| \sum_{s \in \mathcal{S}_k(t)} (D_s - D_k) \Psi_s \right| \leq L_D \varepsilon_{\theta, \text{eff}} \Lambda_k(t).$$

Combining this with (EC.2) yields

$$|M_k(t) - D_k \Lambda_k(t)| \leq C_1 \sqrt{\Lambda_k(t) \log(eKT/\zeta)} + C_2 \log(eKT/\zeta) + L_D \varepsilon_{\theta, \text{eff}} \Lambda_k(t). \quad (\text{EC.3})$$

We now compare the estimated exposure $E_k(t)$ to the true exposure $\Lambda_k(t)$. Since

$$\Psi_s \in [\pi_{\min}, 1],$$

and the clipping interval $[\pi_{\min}/2, 1]$ contains Ψ_s , projection onto this interval cannot increase distance to Ψ_s . Therefore, on the timing-confidence event,

$$|e_s - \Psi_s| \leq \left| \widehat{\Psi}_{s-1}(x_s, p_s, \mu_s) - \Psi_s \right| \leq \varepsilon_\psi. \quad (\text{EC.4})$$

Let

$$m_k(t) := |\mathcal{S}_k(t)|.$$

Then

$$|E_k(t) - \Lambda_k(t)| \leq \varepsilon_\psi m_k(t). \quad (\text{EC.5})$$

Moreover, clipping gives

$$E_k(t) = \sum_{s \in \mathcal{S}_k(t)} e_s \geq \frac{\pi_{\min}}{2} m_k(t), \quad (\text{EC.6})$$

and since $\Psi_s \leq 1$,

$$\Lambda_k(t) \leq m_k(t) \leq \frac{2}{\pi_{\min}} E_k(t). \quad (\text{EC.7})$$

Combining (EC.5) and (EC.6) gives

$$\frac{|E_k(t) - \Lambda_k(t)|}{\max\{1, E_k(t)\}} \leq \frac{2\varepsilon_\psi}{\pi_{\min}}. \quad (\text{EC.8})$$

It remains to control the ratio. Since projection onto $[0, 1]$ cannot increase the distance to $D_k \in [0, 1]$,

$$\left| \widehat{D}_k(t) - D_k \right| \leq \left| \frac{M_k(t)}{\max\{1, E_k(t)\}} - D_k \right|.$$

Let

$$L_t := \log(eKT/\zeta).$$

If $\max\{1, E_k(t)\} \leq 4L_t$, then the desired bound is immediate after increasing c_1 , because the left-hand side is at most one and

$$\sqrt{\frac{L_t}{\max\{1, E_k(t)\}}} \geq \frac{1}{2}.$$

Now suppose $\max\{1, E_k(t)\} > 4L_t$. Then $E_k(t) > 0$, and

$$\left| \frac{M_k(t)}{E_k(t)} - D_k \right| \leq \frac{|M_k(t) - D_k \Lambda_k(t)|}{E_k(t)} + D_k \frac{|\Lambda_k(t) - E_k(t)|}{E_k(t)}.$$

Using (EC.3), (EC.7), and (EC.8), we get

$$\left| \frac{M_k(t)}{E_k(t)} - D_k \right| \leq C \sqrt{\frac{L_t}{E_k(t)}} + C \frac{L_t}{E_k(t)} + C \varepsilon_{\theta, \text{eff}} + C \varepsilon_{\psi}.$$

In the present case $E_k(t) > 4L_t$, the term $L_t/E_k(t)$ is dominated by the square-root term. Hence

$$\left| \frac{M_k(t)}{E_k(t)} - D_k \right| \leq C' \sqrt{\frac{L_t}{E_k(t)}} + C' \varepsilon_{\theta, \text{eff}} + C' \varepsilon_{\psi}.$$

Combining this with the small-exposure case proves that, uniformly over all $k \in [K]$ and $t \leq T$,

$$\left| \widehat{D}_k(t) - D_k \right| \leq c_1 \sqrt{\frac{\log(eKT/\zeta)}{\max\{1, E_k(t)\}}} + c_2 \varepsilon_{\theta, \text{eff}} + c_3 \varepsilon_{\psi}.$$

This completes the proof. $\square$

Proof of Lemma 3. Fix a Mode 0 round t , a feasible offset $k \in \mathcal{A}_t$, and a deadline μ_ℓ . Write

$$p_{t,k} = x_t^\top \widehat{\theta}_{t-1} + \delta_k,$$

and define

$$\begin{aligned} \Psi_{t,k,\ell} &:= \Psi(x_t, p_{t,k}, \mu_\ell), & \widehat{\Psi}_{t,k,\ell} &:= \widehat{\Psi}_{t-1}(x_t, p_{t,k}, \mu_\ell), \\ C_{t,k,\ell} &:= C(x_t, p_{t,k}, \mu_\ell), & \widehat{C}_{t,k,\ell} &:= \widehat{C}_{t-1}(x_t, p_{t,k}, \mu_\ell). \end{aligned}$$

Also let

$$D_{t,k} := D(p_{t,k} - x_t^\top \theta), \quad D_k := D(\delta_k).$$

On a Mode 0 round, both uncertainty gates have passed. On the good event from Lemmas 1 and 2,

$$\left| x_t^\top (\widehat{\theta}_{t-1} - \theta) \right| \leq \varepsilon_{\theta, \text{eff}},$$

and

$$\left| \widehat{\Psi}_{t,k,\ell} - \Psi_{t,k,\ell} \right| \leq \varepsilon_\psi, \quad \left| \widehat{C}_{t,k,\ell} - C_{t,k,\ell} \right| \leq \varepsilon_\psi.$$

Therefore, by Lipschitz continuity of D ,

$$|D_{t,k} - D_k| = \left| D(\delta_k + x_t^\top (\widehat{\theta}_{t-1} - \theta)) - D(\delta_k) \right| \leq L_D \varepsilon_{\theta, \text{eff}}.$$

By Lemma EC.2, on the offset-demand confidence event,

$$\left| \widehat{D}_k(t) - D_k \right| \leq c_1 \sqrt{\frac{\log(eKT/\zeta)}{\max\{1, E_k(t)\}}} + c_2 \varepsilon_{\theta, \text{eff}} + c_3 \varepsilon_\psi.$$

Enlarging constants and applying a union bound over deadlines allows $\log(eKT/\zeta)$ to be replaced by $\log(KLT/\zeta)$.

We now decompose the profit-estimation error. Using $p_{t,k} \leq \bar{p}$, $D_k \in [0, 1]$, and $\Psi_{t,k,\ell} \in [0, 1]$,

$$\begin{aligned} & \left| p_{t,k} \widehat{D}_k(t) \widehat{\Psi}_{t,k,\ell} - p_{t,k} D_{t,k} \Psi_{t,k,\ell} \right| \\ & \leq \bar{p} \left| \widehat{D}_k(t) \widehat{\Psi}_{t,k,\ell} - D_{t,k} \Psi_{t,k,\ell} \right| \\ & \leq \bar{p} \widehat{\Psi}_{t,k,\ell} \left| \widehat{D}_k(t) - D_k \right| + \bar{p} D_k \left| \widehat{\Psi}_{t,k,\ell} - \Psi_{t,k,\ell} \right| + \bar{p} \Psi_{t,k,\ell} |D_k - D_{t,k}|. \end{aligned}$$

Substituting the offset-demand confidence bound, the timing error bound, and the current valuation-centering bound gives

$$\begin{aligned} & \left| p_{t,k} \widehat{D}_k(t) \widehat{\Psi}_{t,k,\ell} - p_{t,k} D_{t,k} \Psi_{t,k,\ell} \right| \\ & \leq \kappa_2 \widehat{\Psi}_{t,k,\ell} \sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_k(t)\}}} + \kappa'_3 \varepsilon_{\theta, \text{eff}} + \kappa'_4 \varepsilon_\psi \end{aligned}$$

for constants depending only on the operating-domain constants.

The holding-cost term satisfies

$$\bar{h} \left| \widehat{C}_{t,k,\ell} - C_{t,k,\ell} \right| \leq \bar{h} \varepsilon_\psi,$$

which is absorbed into the ε_ψ term.

Finally, COPDE includes the nonnegative pending-window correction

$$\kappa_5 \frac{\bar{\mu}}{\max\{1, N_k(t)\}}.$$

This term accounts for the difference between the sent-offer count $N_k(t)$ and the matured statistics $M_k(t), E_k(t)$. Since every deadline is at most $\bar{\mu}$, at most $\bar{\mu}$ previously sent offset- k offers can remain unmatured at any calendar time. Adding this correction yields

$$\begin{aligned} & |\widehat{r}_t(k, \ell) - r(x_t; p_{t,k}, \mu_\ell)| \\ & \leq \kappa_2 \widehat{\Psi}_{t-1}(x_t, p_{t,k}, \mu_\ell) \sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_k(t)\}}} + \kappa_5 \frac{\bar{\mu}}{\max\{1, N_k(t)\}} + \kappa_3 \varepsilon_{\theta, \text{eff}} + \kappa_4 \varepsilon_\psi. \end{aligned}$$

This proves the lemma. $\square$

EC.4.3. Regret Analysis

LEMMA EC.3. Fix a Mode 0 round t . For $k \in \mathcal{A}_t$ and $\ell \in [L]$, write

$$v_t(k, \ell) := r(x_t; p_{t,k}, \mu_\ell), \quad p_{t,k} := x_t^\top \widehat{\theta}_{t-1} + \delta_k.$$

Suppose the good event of Lemma 3 holds:

$$|v_t(k, \ell) - \widehat{r}_t(k, \ell)| \leq b_t(k, \ell) \quad \forall k \in \mathcal{A}_t, \ell \in [L],$$

where $b_t(k, \ell)$ is the radius in (16). Define

$$\bar{v}_t(k) := \max_{\ell \in [L]} v_t(k, \ell), \quad \bar{b}_t(k) := \max_{\ell \in [L]} b_t(k, \ell).$$

Let

$$k_t^\star \in \arg \max_{k \in \mathcal{A}_t} \bar{v}_t(k)$$

be the current grid-oracle offset. Then $k_t^\star \in \mathcal{K}_t$. Moreover, if COPDE chooses

$$k_t \in \arg \min_{k \in \mathcal{K}_t} N_k(t), \quad \ell_t \in \arg \max_{\ell \in [L]} \text{UCB}_t(k_t, \ell),$$

then

$$\bar{v}_t(k_t^\star) - v_t(k_t, \ell_t) \leq 2\bar{b}_t(k_t^\star) + 4\bar{b}_t(k_t).$$

Consequently,

$$\bar{v}_t(k_t^\star) - v_t(k_t, \ell_t) \leq C \left[\sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_{k_t}(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_{k_t}(t)\}} + \varepsilon_{\theta, \text{eff}} + \varepsilon_\psi \right],$$

for a constant C depending only on the operating-domain constants.

Proof. For every pair (k, ℓ) , the good event implies

$$\text{LCB}_t(k, \ell) \leq v_t(k, \ell) \leq \text{UCB}_t(k, \ell).$$

Taking maxima over ℓ gives

$$\text{LCB}_t(k) = \max_{\ell} \text{LCB}_t(k, \ell) \leq \bar{v}_t(k) \leq \max_{\ell} \text{UCB}_t(k, \ell) = \text{UCB}_t(k).$$

Since k_t^* maximizes $\bar{v}_t(k)$ over $\mathcal{A}_t$, for every $j \in \mathcal{A}_t$,

$$\text{UCB}_t(k_t^*) \geq \bar{v}_t(k_t^*) \geq \bar{v}_t(j) \geq \text{LCB}_t(j).$$

Thus

$$\text{UCB}_t(k_t^*) \geq \max_{j \in \mathcal{A}_t} \text{LCB}_t(j),$$

and therefore $k_t^* \in \mathcal{K}_t$.

We next compare the selected offset with the grid-oracle offset. Since $k_t \in \mathcal{K}_t$,

$$\text{UCB}_t(k_t) \geq \max_{j \in \mathcal{A}_t} \text{LCB}_t(j) \geq \text{LCB}_t(k_t^*).$$

Hence

$$\bar{v}_t(k_t^*) - \bar{v}_t(k_t) \leq (\bar{v}_t(k_t^*) - \text{LCB}_t(k_t^*)) + (\text{UCB}_t(k_t) - \bar{v}_t(k_t)).$$

For any offset k , let $\ell^v(k) \in \arg \max_{\ell} v_t(k, \ell)$. Then

$$\text{LCB}_t(k) \geq \text{LCB}_t(k, \ell^v(k)) \geq v_t(k, \ell^v(k)) - 2b_t(k, \ell^v(k)) = \bar{v}_t(k) - 2b_t(k, \ell^v(k)),$$

so

$$\bar{v}_t(k) - \text{LCB}_t(k) \leq 2\bar{b}_t(k).$$

Similarly, if $\ell^U(k) \in \arg \max_{\ell} \text{UCB}_t(k, \ell)$, then

$$\text{UCB}_t(k) = \text{UCB}_t(k, \ell^U(k)) \leq v_t(k, \ell^U(k)) + 2b_t(k, \ell^U(k)) \leq \bar{v}_t(k) + 2\bar{b}_t(k),$$

so

$$\text{UCB}_t(k) - \bar{v}_t(k) \leq 2\bar{b}_t(k).$$

Therefore,

$$\bar{v}_t(k_t^*) - \bar{v}_t(k_t) \leq 2\bar{b}_t(k_t^*) + 2\bar{b}_t(k_t).$$

It remains to account for the deadline chosen after the offset. Let

$$\ell_t^v \in \arg \max_{\ell \in [L]} v_t(k_t, \ell).$$

Since $\ell_t \in \arg \max_{\ell} \text{UCB}_t(k_t, \ell)$,

$$\text{UCB}_t(k_t, \ell_t) \geq \text{UCB}_t(k_t, \ell_t^v).$$

Thus

$$\bar{v}_t(k_t) = v_t(k_t, \ell_t^v) \leq \text{UCB}_t(k_t, \ell_t^v) \leq \text{UCB}_t(k_t, \ell_t) \leq v_t(k_t, \ell_t) + 2b_t(k_t, \ell_t) \leq v_t(k_t, \ell_t) + 2\bar{b}_t(k_t).$$

Combining the last two displays gives

$$\bar{v}_t(k_t^*) - v_t(k_t, \ell_t) \leq 2\bar{b}_t(k_t^*) + 4\bar{b}_t(k_t).$$

Finally, since both k_t^* and k_t belong to $\mathcal{K}_t$, and COPDE selects an offset with minimal $N_k(t)$ in $\mathcal{K}_t$,

$$N_{k_t}(t) \leq N_{k_t^*}(t).$$

For any offset k , at most $\bar{\mu}$ sent offset- k offers are pending at time t . Hence the number of matured offset- k offers is at least

$$(N_k(t) - \bar{\mu})_+.$$

By the clipping rule,

$$E_k(t) \geq \frac{\pi_{\min}}{2} (N_k(t) - \bar{\mu})_+.$$

Therefore,

$$\sqrt{\frac{1}{\max\{1, E_k(t)\}}} \leq C \left[\sqrt{\frac{1}{\max\{1, N_k(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_k(t)\}} \right].$$

Since $N_{k_t}(t) \leq N_{k_t^*}(t)$, the N -based bound for the grid-oracle offset is no larger than the corresponding bound for the selected offset, up to constants. Substituting the definition of $b_t(k, \ell)$, using $\hat{\Psi}_{t-1} \leq 1$, and absorbing constants gives

$$\bar{b}_t(k_t^*) \leq C \left[\sqrt{\frac{\log(KLT/\zeta)}{\max\{1, E_{k_t}(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_{k_t}(t)\}} + \varepsilon_{\theta, \text{eff}} + \varepsilon_{\psi} \right],$$

and the same bound holds directly for $\bar{b}_t(k_t)$. Combining this with

$$\bar{v}_t(k_t^*) - v_t(k_t, \ell_t) \leq 2\bar{b}_t(k_t^*) + 4\bar{b}_t(k_t)$$

proves the final display. □

LEMMA EC.4. For any integer $N \geq 1$, $a > 0$, $b \geq 0$, and $\alpha > 0$,

$$\sum_{n=1}^N \left(\frac{a}{\sqrt{n}} + \frac{b}{n} \right) \leq \frac{a^2}{\alpha} + \alpha N + b \log(eN).$$

Proof. Using

$$\sum_{n=1}^N \frac{1}{\sqrt{n}} \leq 2\sqrt{N}$$

and

$$2a\sqrt{N} \leq \frac{a^2}{\alpha} + \alpha N,$$

we obtain

$$\sum_{n=1}^N \frac{a}{\sqrt{n}} \leq 2a\sqrt{N} \leq \frac{a^2}{\alpha} + \alpha N.$$

Also,

$$\sum_{n=1}^N \frac{1}{n} \leq \log(eN).$$

Combining the two bounds proves the claim. $\square$

LEMMA EC.5. Under Assumption 1 and the profit factorization (8), there exists a constant $L_r < \infty$, depending only on the operating-domain constants, such that for every fixed $x \in X$, every $\mu \in \mathcal{M}$, and all $p, p' \in [\underline{p}, \bar{p}]$,

$$|r(x; p, \mu) - r(x; p', \mu)| \leq L_r |p - p'|.$$

Proof. Fix $x \in X$ and $\mu \in \mathcal{M}$. Write

$$D_p := D(p - x^\top \theta), \quad \Psi(p) := \Psi(x, p, \mu), \quad C(p) := C(x, p, \mu).$$

By Lipschitzness of D ,

$$|D_p - D_{p'}| \leq L_D |p - p'|.$$

Moreover, since

$$\Psi(p) = g_\mu(h(x, p, \mu)^\top \psi),$$

the slope bound on g_μ , the parameter bound $\|\psi\| \leq B$, and the Lipschitzness of $p \mapsto h(x, p, \mu)$ give

$$|\Psi(p) - \Psi(p')| \leq L_g |h(x, p, \mu)^\top \psi - h(x, p', \mu)^\top \psi| \leq L_g B L_h |p - p'|.$$

Similarly,

$$|C(p) - C(p')| \leq L_C B L_h |p - p'|.$$

Using $D_p, D_{p'}, \Psi(p), \Psi(p') \in [0, 1]$ and $p, p' \in [\underline{p}, \bar{p}]$, we have

$$\begin{aligned} & |pD_p\Psi(p) - p'D_{p'}\Psi(p')| \\ & \leq |p - p'|D_p\Psi(p) + p'|D_p - D_{p'}|\Psi(p) + p'D_{p'}|\Psi(p) - \Psi(p')| \\ & \leq (1 + \bar{p}L_D + \bar{p}L_gBL_h)|p - p'|. \end{aligned}$$

The holding-cost term satisfies

$$\bar{h}|C(p) - C(p')| \leq \bar{h}L_CBL_h|p - p'|.$$

Combining the revenue and open-time terms gives

$$|r(x; p, \mu) - r(x; p', \mu)| \leq (1 + \bar{p}L_D + \bar{p}L_gBL_h + \bar{h}L_CBL_h)|p - p'|.$$

Thus the claim holds with, for example,

$$L_r := 1 + \bar{p}L_D + \bar{p}L_gBL_h + \bar{h}L_CBL_h.$$

□

Proof of Theorem 1. Let

$$G_2 := \left\{ t \leq T : \max_{(p, \mu) \in \mathcal{P} \times \mathcal{M}} \|h(x_t, p, \mu)\|_{V_{h, t-1}^{-1}} > \eta_h \right\}$$

be the set of timing-exploration rounds, and let

$$G_1 := \left\{ t \leq T : t \notin G_2, \|x_t\|_{V_{t-1}^{-1}} > \eta \right\}$$

be the set of valuation-exploration rounds. This ordering matches Algorithm 1: timing uncertainty is checked before valuation uncertainty because Mode 1 uses an IPW correction based on the timing model.

Let $\mathcal{E}$ denote the intersection of the high-probability events in Lemmas 2, 1, and 3. By a union bound with $\zeta_T = 1/(6T)$, $\mathbb{P}(\mathcal{E}) \geq 1 - O(T^{-1})$. We prove the regret bound on $\mathcal{E}$.

The regret decomposes as

$$\begin{aligned} \text{Regret}_T(\pi) &= \sum_{t=1}^T [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)] \\ &= \sum_{t \in G_2} [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)] \end{aligned} \tag{EC.9}$$

$$+ \sum_{t \in G_1} [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)] \tag{EC.10}$$

$$+ \sum_{t \notin G_1 \cup G_2} [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)]. \tag{EC.11}$$

Let

$$R_{\max} := \bar{p} + \bar{h} \bar{\mu}.$$

Since each per-period regret is bounded by $R_{\max}$, Lemma EC.1 gives

$$|G_1| \leq \frac{2(1 + \bar{\mu})d}{\eta^2} \log \left(1 + \frac{TB^2}{\lambda d} \right),$$

and

$$|G_2| \leq \frac{2(1 + \bar{\mu})d'}{\underline{\kappa}_\ell \eta_h^2} \log \left(1 + \frac{T \underline{\kappa}_\ell B_h^2}{\lambda d'} \right).$$

Therefore,

$$\sum_{t \in G_1 \cup G_2} [r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t)] \leq R_{\max}(|G_1| + |G_2|) = \tilde{O}(T^{2/3}),$$

under the theorem's choices of η and η_h .

We now consider a Mode 0 round $t \notin G_1 \cup G_2$. Since $t \notin G_2$, Lemma 2 and the definition of η_h imply

$$\left| \widehat{\Psi}_{t-1}(x_t, p, \mu) - \Psi(x_t, p, \mu) \right| \leq \varepsilon_\psi, \quad \left| \widehat{C}_{t-1}(x_t, p, \mu) - C(x_t, p, \mu) \right| \leq \varepsilon_\psi$$

for every $(p, \mu) \in \mathcal{P} \times \mathcal{M}$. Moreover, each Mode 1 round $s \in G_1$ satisfies $s \notin G_2$. The timing gate controls the maximum leverage over the full price–deadline domain before the probe price is drawn. Lemma 2 therefore gives

$$\sup_{p \in [\underline{p}, \bar{p}]} \left| \widehat{\Psi}_{s-1}(x_s, p, \bar{\mu}) - \Psi(x_s, p, \bar{\mu}) \right| \leq \varepsilon_\psi.$$

Since $\Psi(x_s, p, \bar{\mu}) \in [\pi_{\min}, 1]$ for every such price, clipping cannot increase the error. Thus

$$\sup_{p \in [\underline{p}, \bar{p}]} \left| \widehat{\Psi}_{s-1}^c(x_s, p, \bar{\mu}) - \Psi(x_s, p, \bar{\mu}) \right| \leq \varepsilon_\psi,$$

which establishes $\mathcal{E}_\psi^\theta(\varepsilon_\psi)$, the uniform Mode 1 timing-good event required by Lemma 1.

Since $t \notin G_1$, we have

$$\|x_t\|_{V_{t-1}^{-1}} \leq \eta.$$

Lemma 1 then gives

$$|x_t^\top (\widehat{\theta}_{t-1} - \theta)| \leq \beta_T^{\theta, \text{ipw}}(\zeta_T) \eta + C_{\text{ipw}} \varepsilon_\psi \sqrt{N_{t-1}^\theta} \eta.$$

The first term equals ε_θ . Also, $N_{t-1}^\theta \leq |G_1|$, so the exploration-count bound implies

$$N_{t-1}^\theta \eta^2 \leq Cd \log T$$

for a constant C depending only on the operating-domain constants. Hence

$$C_{\text{ipw}} \varepsilon_\psi \sqrt{N_{t-1}^\theta} \eta \leq C_{\theta\psi} \sqrt{d \log T} \varepsilon_\psi.$$

By the definition of $\varepsilon_{\theta, \text{eff}}$,

$$|x_t^\top (\widehat{\theta}_{t-1} - \theta)| \leq \varepsilon_{\theta, \text{eff}}. \quad (\text{EC.12})$$

For each feasible offset $k \in \mathcal{A}_t$, define

$$p_{t,k} := x_t^\top \widehat{\theta}_{t-1} + \delta_k$$

and

$$\bar{v}_t(k) := \max_{\ell \in [L]} r(x_t; p_{t,k}, \mu_\ell).$$

Let

$$k_t^\star \in \arg \max_{k \in \mathcal{A}_t} \bar{v}_t(k).$$

We decompose the Mode 0 regret as

$$\begin{aligned} & r(x_t; p^*(x_t), \mu^*(x_t)) - r(x_t; p_t, \mu_t) \\ &= \underbrace{r(x_t; p^*(x_t), \mu^*(x_t)) - \bar{v}_t(k_t^\star)}_{\text{discretization}} + \underbrace{\bar{v}_t(k_t^\star) - r(x_t; p_t, \mu_t)}_{\text{offset elimination}}. \end{aligned}$$

For the discretization term, let

$$\delta_t^\star := p^*(x_t) - x_t^\top \theta.$$

Since the offset grid discretizes the feasible operating offset set, there exists a feasible offset $k \in \mathcal{A}_t$ such that

$$|\delta_k - \delta_t^\star| \leq \eta_\Delta.$$

For that offset,

$$|p_{t,k} - p^*(x_t)| = \left| x_t^\top (\widehat{\theta}_{t-1} - \theta) + \delta_k - \delta_t^\star \right| \leq \varepsilon_{\theta, \text{eff}} + \eta_\Delta.$$

Using the oracle deadline $\mu^*(x_t)$ for comparison and applying Lemma EC.5,

$$r(x_t; p^*(x_t), \mu^*(x_t)) - \bar{v}_t(k_t^\star) \leq L_r(\eta_\Delta + \varepsilon_{\theta, \text{eff}}). \quad (\text{EC.13})$$

Summing (EC.13) over $t \leq T$ gives

$$O(T\eta_\Delta + T\varepsilon_{\theta, \text{eff}}) = \widetilde{O}(T^{2/3}).$$

For the offset-elimination term, Lemma 3 and the elimination rule imply that the grid-oracle offset $k_t^\star$ is retained in $\mathcal{K}_t$. Since COPDE selects the least-sampled surviving offset and then an optimistic deadline, Lemma EC.3 yields

$$\bar{v}_t(k_t^\star) - r(x_t; p_t, \mu_t) \leq C \left[\widehat{\Psi}_{t-1}(x_t, p_t, k_t, \mu_{\ell_t}) \sqrt{\frac{\log(KLT/\zeta_T)}{\max\{1, E_{k_t}(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_{k_t}(t)\}} + \varepsilon_{\theta, \text{eff}} + \varepsilon_\psi \right]. \quad (\text{EC.14})$$

We now sum (EC.14) over Mode 0 rounds. Since $\widehat{\Psi}_{t-1} \leq 1$, the first term is bounded by

$$\sqrt{\frac{\log(KLT/\zeta_T)}{\max\{1, E_{k_t}(t)\}}}.$$

For each offset k , at most $\bar{\mu}$ previously posted offset- k offers can be pending at any time. By the clipping rule and the lower bound on deadline-meeting probabilities,

$$E_k(t) \geq \frac{\pi_{\min}}{2} (N_k(t) - \bar{\mu})_+.$$

Therefore, for a constant $C < \infty$,

$$\sqrt{\frac{1}{\max\{1, E_k(t)\}}} \leq C \left[\frac{1}{\sqrt{\max\{1, N_k(t)\}}} + \frac{\bar{\mu}}{\max\{1, N_k(t)\}} \right].$$

Hence the Mode 0 per-round regret is bounded, up to constants, by

$$\frac{a}{\sqrt{\max\{1, N_{k_t}(t)\}}} + \frac{b}{\max\{1, N_{k_t}(t)\}} + \varepsilon_{\theta, \text{eff}} + \varepsilon_\psi,$$

where

$$a \asymp \sqrt{\log(KLT/\zeta_T)}, \quad b \asymp \bar{\mu}.$$

For each offset k , enumerate the Mode 0 rounds on which k is selected. Applying Lemma EC.4 to the selected-offset sequence gives, for any $\alpha > 0$,

$$\sum_{\substack{t \notin G_1 \cup G_2: \\ k_t = k}} [\bar{v}_t(k_t^\star) - r(x_t; p_t, \mu_t)] \leq \tilde{O} \left(\frac{1}{\alpha} + \alpha N_k(T+1) + \bar{\mu} \log T + N_k(T+1)(\varepsilon_{\theta, \text{eff}} + \varepsilon_\psi) \right).$$

Summing over $k \in [K]$ and using

$$\sum_{k=1}^K N_k(T+1) \leq T$$

yields

$$\sum_{t \notin G_1 \cup G_2} [\bar{v}_t(k_t^*) - r(x_t; p_t, \mu_t)] \leq \tilde{O}(K/\alpha + T\alpha + T\varepsilon_{\theta, \text{eff}} + T\varepsilon_{\psi} + K\bar{\mu}). \quad (\text{EC.15})$$

Taking $\alpha = T^{-1/3}$ and $K = O(T^{1/3})$, the right-hand side of (EC.15) is

$$\tilde{O}(T^{2/3}).$$

Combining the exploration, discretization, and offset-elimination bounds proves

$$\text{Regret}_T(\pi) = \tilde{O}(T^{2/3})$$

on the event $\mathcal{E}$. Since $\mathbb{P}(\mathcal{E}) \geq 1 - O(T^{-1})$, the theorem follows. $\square$

EC.5. Proofs for Rate Optimality and Deadline Identification

Proof of Theorem 2. We reduce from the standard non-contextual dynamic-pricing problem with i.i.d. valuations $v_t \in [0, 1]$ and an unknown Lipschitz survival function $Q(p) := \mathbb{P}(v_t \geq p)$. In that problem, a seller chooses a price $p_t \in [0, 1]$, observes $y_t = \mathbb{I}\{v_t \geq p_t\}$, and receives expected revenue

$$R_Q(p_t) = p_t Q(p_t).$$

We establish the required pricing lower bound directly. Define the baseline survival function

$$Q_0(p) = \begin{cases} 1, & p \leq 1/4, \\ 1/(4p), & 1/4 < p \leq 3/4, \\ (4/3)(1-p), & 3/4 < p \leq 1, \\ 0, & p > 1. \end{cases}$$

Its revenue $R_0(p) = pQ_0(p)$ is at most $1/4$ on $[0, 1]$, with equality throughout $[1/4, 3/4]$. For any integer $K \geq 2$, set $\ell = 1/(4K)$ and $H = \ell/16 = 1/(64K)$. Let

$$J_j = (3/8 + (j-1)\ell, 3/8 + j\ell), \quad z_j = 3/8 + (j-1/2)\ell, \quad j = 1, \dots, K,$$

and define triangular revenue perturbations

$$b_j(p) = H \left(1 - \frac{2|p - z_j|}{\ell} \right)_+, \quad Q_j(p) = \begin{cases} Q_0(p) + b_j(p)/p, & p \in J_j, \\ Q_0(p), & p \notin J_j, \end{cases}$$

where $(a)_+ = \max\{a, 0\}$. Each Q_j is continuous. On J_j , its derivative, wherever it exists, satisfies

$$Q'_j(p) = \frac{-1/4 + pb'_j(p) - b_j(p)}{p^2} < 0, \quad |Q'_j(p)| \leq \frac{1/4 + 5/64 + 1/128}{(3/8)^2} = \frac{43}{18} < 4,$$

because $|b'_j| = 1/8$, $p \in [3/8, 5/8]$, and $H \leq 1/128$. Outside J_j , the slope of Q_0 has magnitude at most 4. Thus these are survival functions of atomless distributions supported on $[0, 1]$, all globally 4-Lipschitz. Let $\mathcal{Q}$ contain Q_0 and all these alternatives for every $K \geq 2$; this family and its regularity constants do not depend on the horizon.

Fix any horizon $T \geq 1$ and any possibly randomized adaptive pricing policy, and choose $K = \max\{2, \lceil T^{1/3} \rceil\}$. Let $\mathbb{P}_j$ be the joint law of its prices and binary purchase observations under Q_j , and write

$$N_j = \sum_{t=1}^T \mathbb{I}\{p_t \in J_j\}, \quad n_j = \mathbb{E}_0[N_j].$$

The intervals J_j are disjoint, so $\sum_j n_j \leq T$. On J_j , both purchase probabilities belong to $[1/4, 3/4]$, and $|Q_j(p) - Q_0(p)| \leq 4H$; outside J_j they coincide. The Bernoulli relative entropy bound $\text{kl}(q, q') \leq (q - q')^2 / [q'(1 - q')]$ and the chain rule therefore give

$$\text{KL}(\mathbb{P}_0 \parallel \mathbb{P}_j) \leq 128H^2 n_j.$$

The price-selection kernels are the same under all instances, so this bound applies to adaptive and randomized policies. Since $0 \leq N_j \leq T$, Pinsker's inequality yields

$$\mathbb{E}_j[N_j] \leq n_j + 8TH\sqrt{n_j}.$$

Averaging and using Cauchy–Schwarz, $K \geq 2$, and $K^3 \geq T$,

$$\frac{1}{K} \sum_{j=1}^K \mathbb{E}_j[N_j] \leq \frac{T}{K} + 8TH\sqrt{\frac{T}{K}} \leq \frac{5T}{8}.$$

Under Q_j , optimal expected revenue is $1/4 + H$, attained at z_j , whereas every price outside J_j earns at most $1/4$. Hence

$$\begin{aligned} \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\text{Regret}_T^{\text{std}}] &\geq \frac{1}{K} \sum_{j=1}^K H(T - \mathbb{E}_j[N_j]) \\ &\geq \frac{3HT}{8} = \frac{3T}{512K} \geq \frac{3}{1024} T^{2/3}, \end{aligned}$$

where the last inequality uses $K \leq 2T^{1/3}$. This proves the required bound with $c_0 = 3/1024$, uniformly over policies and horizons.

For each Q in this family, let $m_Q := \mathbb{E}_Q[v_t]$. Embed the valuation distribution using a fixed scalar context $x_t \equiv 1$ and set

$$\theta_Q := m_Q, \quad \xi_t := v_t - m_Q.$$

Then the shocks are i.i.d. with $\mathbb{E}_Q[\xi_t] = 0$, while

$$u_t = x_t \theta_Q + \xi_t = v_t \in [0, 1] \quad \text{almost surely.}$$

Moreover, $|\theta_Q| \leq 1$ and $|\xi_t| \leq 1$ uniformly over the family. Writing $F_{\xi, Q}$ for the CDF of the centered shock, the residual demand function in the price–deadline model is

$$D_Q(z) := 1 - F_{\xi, Q}(z) = Q(z + m_Q).$$

Translation preserves the Lipschitz constant, and

$$D_Q(p - x_t \theta_Q) = D_Q(p - m_Q) = Q(p).$$

Thus the embedding preserves every purchase probability while satisfying the bounded-valuation and zero-mean-shock conditions. The mean m_Q and the residual demand function D_Q remain unknown to the learner.

Set $\mathcal{P} = [0, 1]$, $\underline{p} = 0$, $\bar{p} = 1$, and take a single admissible deadline $\mu_0 = 1$. Independently of the valuations, draw $d_t = 1$ with probability $\rho \in (0, 1)$ and $d_t = 2$ otherwise, independently across periods. This known timing process satisfies

$$\Psi(1, p, \mu_0) = \rho$$

and

$$C(1, p, \mu_0) = 1.$$

This subclass satisfies the timing regularity conditions. For example, take $h(x, p, \mu_0) \equiv 1$, a logistic deadline-meeting link $g_{\mu_0}(z) = (1 + e^{-z})^{-1}$, and the fixed parameter $\psi_0 = \log(\rho/(1 - \rho))$, so that $g_{\mu_0}(\psi_0) = \rho$. The Bernoulli timing loss has uniformly bounded score and positive curvature on any bounded operating domain around ψ_0 . The open-time function may be taken as $c_{\mu_0}(z) \equiv 1$, which is differentiable with bounded slope.

Since there is only one deadline, the open-time term $\bar{h}$ is a constant and cancels from regret. The expected profit in this subclass is

$$r_Q(p, \mu_0) = \rho p D_Q(p - m_Q) - \bar{h} = \rho p Q(p) - \bar{h}.$$

The oracle price is the same as in the standard pricing problem, and the per-period regret is exactly ρ times the standard pricing regret.

Fix any admissible policy π for the price–deadline problem. We construct a policy π^{std} for the standard non-contextual pricing problem. At each round, π^{std} runs the policy π , posts the same price p_t , observes the standard purchase label y_t , and independently simulates a deadline-meeting indicator $w_t \sim \text{Bernoulli}(\rho)$. It stores the simulated feedback and releases it on the embedded instance’s deadline-maturity schedule: if $w_t = 1$, the label y_t is revealed; if $w_t = 0$, the label is censored and $B_t = w_t y_t = 0$. This simulation requires neither m_Q nor D_Q . Therefore, the internal history of π under this simulation has the same distribution as in the embedded deadline instance.

For every demand curve Q , the expected regret of π in the embedded deadline instance equals ρ times the expected regret of π^{std} in the standard pricing instance:

$$\mathbb{E}_Q [\text{Regret}_T^{\text{deadline}}(\pi)] = \rho \mathbb{E}_Q [\text{Regret}_T^{\text{std}}(\pi^{\text{std}})].$$

Let $\mathfrak{I}$ consist of the embeddings of $Q \in \mathcal{Q}$ above, with ρ and all operating-domain constants fixed independently of T . Taking the supremum over $Q \in \mathcal{Q}$ and applying the pricing lower bound to π^{std} gives

$$\sup_{I \in \mathfrak{I}} \mathbb{E}_I [\text{Regret}_T(\pi)] \geq \rho c_0 T^{2/3}.$$

Thus $c = \rho c_0 > 0$ works for every horizon and every admissible policy. □

Proof of Proposition 1. We construct two instances with a single context $x_t \equiv 1$, a fixed price $\mathcal{P} = \{p_0\}$ with $p_0 = 1/2$, and deadlines $\mathcal{M} = \{1, 2\}$. Set $\underline{p} = 0$, $\bar{p} = 1$, and let

$$q_0 = \frac{1}{2}, \quad a = \frac{1}{4}, \quad b = \frac{3}{8}, \quad s(z) = \frac{2}{1 + e^{-z}}.$$

The function s is strictly increasing, with $s(0) = 1$ and $s'(0) = 1/2$. Choose a sufficiently small constant $\epsilon_0 \in (0, 1)$ such that, for all $|z| \leq \epsilon_0$,

$$0 < as(z) < bs(z) < 1, \quad q_0/s(z) \in (0, 1).$$

The constant ϵ_0 is also chosen to satisfy the curvature condition verified below. Fix any $\epsilon \in (0, \epsilon_0]$, independently of the horizon. For $\sigma \in \{-1, +1\}$, define

$$z_\sigma = \sigma \epsilon, \quad s_\sigma = s(z_\sigma), \quad q_\sigma = \frac{q_0}{s_\sigma}.$$

For the valuation model, let the shocks be i.i.d. with $\xi_t \sim \text{Unif}[-\ell_0, \ell_0]$, where $\ell_0 = 1/8$, and set

$$\theta_\sigma = p_0 - \ell_0(1 - 2q_\sigma), \quad u_t = \theta_\sigma + \xi_t.$$

The shocks have zero mean, the valuations lie in $[0, 1]$, and the residual demand function is Lipschitz. Since $q_\sigma \in (0, 1)$,

$$\mathbb{P}_\sigma(y_t = 1) = \mathbb{P}(\xi_t \geq \ell_0(1 - 2q_\sigma)) = q_\sigma.$$

Use the common known timing feature $h(x, p, \mu) \equiv \epsilon_0$ and let $\psi_\sigma = z_\sigma/\epsilon_0$. Then $|\psi_\sigma| \leq 1$ and the true timing index is z_σ . Taking $B = 1$ and $B_h = B_z = \epsilon_0$ ensures that the entire operating index domain in Assumption 1 is $[-\epsilon_0, \epsilon_0]$. Define the common known timing links by

$$g_1(z) = as(z), \quad g_2(z) = bs(z).$$

These links map $\mathbb{R}$ into $(0, 1)$. To verify the loss-curvature condition, write $\gamma_1 = a$, $\gamma_2 = b$. For $w = 1$,

$$\ell_\mu(1, z) = -\log \gamma_\mu - \log s(z)$$

has strictly positive second derivative on every compact interval. For $w = 0$,

$$\ell_\mu(0, z) = -\log(1 - \gamma_\mu s(z)), \quad \partial_{zz}^2 \ell_\mu(0, 0) = \frac{\gamma_\mu^2 (s'(0))^2}{(1 - \gamma_\mu)^2} > 0,$$

where we used $s''(0) = 0$. By continuity, choosing ϵ_0 sufficiently small gives uniformly positive curvature for both outcomes and deadlines on the entire operating domain. The score and link derivatives are bounded there as well.

Independently of the valuation shocks and independently across periods, draw the delay at timing index z according to

$$\mathbb{P}_z(d_t = 1) = as(z), \quad \mathbb{P}_z(d_t = 2) = (b - a)s(z), \quad \mathbb{P}_z(d_t = 3) = 1 - bs(z).$$

This distribution yields

$$\Psi_1^\sigma = as_\sigma, \quad \Psi_2^\sigma = bs_\sigma, \quad C_1^\sigma = 1, \quad C_2^\sigma = 2 - as_\sigma.$$

Thus the common known open-time links are $c_1(z) = 1$ and $c_2(z) = 2 - as(z)$, with bounded derivatives. All conditions of Assumption 1 are therefore satisfied.

Latent demand and deadline meeting cancel in the purchase probabilities:

$$\mathbb{E}_\sigma[B_t \mid \mu_t = 1] = q_\sigma as_\sigma = q_0 a, \quad \mathbb{E}_\sigma[B_t \mid \mu_t = 2] = q_\sigma bs_\sigma = q_0 b.$$

Hence the binary purchase-before-expiry indicator has the same distribution in both instances at each deadline.

Set the common holding-cost rate to

$$\bar{h} = \frac{p_0 q_0 (b - a)}{1 - a}.$$

The expected profits are

$$r^\sigma(1) = p_0 q_0 a - \bar{h}, \quad r^\sigma(2) = p_0 q_0 b - \bar{h}(2 - a s_\sigma).$$

Consequently,

$$r^\sigma(2) - r^\sigma(1) = p_0 q_0 (b - a) - \bar{h}(1 - a s_\sigma) = \bar{h} a (s_\sigma - 1).$$

Since $s_+ > 1 > s_-$, deadline 2 is uniquely optimal in instance +, whereas deadline 1 is uniquely optimal in instance -. The difference arises entirely from expected holding costs, despite identical purchase conversion at each deadline. $\square$

EC.6. Details for Deadline-Augmented Purchase Propensity

This appendix gives the algorithmic changes and proof adaptations for Section 6. Write $\tilde{d} = d + q$, $\phi_{t,\ell} = \phi(x_t, \mu_\ell)$, and $\phi_t = \phi(x_t, \mu_t)$. The timing model and its exploration gate are unchanged.

Conditions for the extension. Impose the counterparts of Assumption 1 on the augmented class: $\|\phi(x, \mu)\| \leq B_\phi$ and $\|\omega\| \leq B_\omega$ uniformly, together with the same demand smoothness, bounded timing features, link regularity, and timing-loss curvature. The valuation support condition must hold uniformly over admissible deadlines:

$$\phi(x, \mu)^\top \omega + \xi \in [\underline{p}, \bar{p}] \quad \text{almost surely for every } (x, \mu) \in \mathcal{X} \times \mathcal{M}.$$

The base stochastic conditions are retained in their sequential form: conditional on the current context and pre-offer history, the valuation shock has the common, zero-mean law defining D , and policy randomization is independent of the fresh customer shocks. Conditional on that history and the posted offer, the timing links remain valid and the valuation and timing shocks remain separated, giving the factorization in (17).

Feasible pairs and Mode 0. For the candidate prices $p_{t,k,\ell}$ in Section 6, define

$$\mathcal{L}_t(k) := \{\ell \in [L] : p_{t,k,\ell} \in [\underline{p}, \overline{p}]\}, \quad \mathcal{A}_t^\dagger := \{k \in [K] : \mathcal{L}_t(k) \neq \emptyset\}.$$

The permanent offset statistics $N_k(t), M_k(t), E_k(t)$ retain their base definitions, using each offer's actual deadline μ_s and the exposure $e_s = \widehat{\Psi}_{s-1}^c(x_s, p_s, \mu_s)$ stored at posting. Thus

$$\widehat{D}_k(t) = \left\lfloor \frac{M_k(t)}{\max\{1, E_k(t)\}} \right\rfloor_{[0,1]},$$

and, for $\ell \in \mathcal{L}_t(k)$, the estimated profit is

$$\widehat{r}_t^\dagger(k, \ell) = p_{t,k,\ell} \widehat{D}_k(t) \widehat{\Psi}_{t-1}(x_t, p_{t,k,\ell}, \mu_\ell) - \bar{h} \widehat{C}_{t-1}(x_t, p_{t,k,\ell}, \mu_\ell).$$

Let $b_t^\dagger(k, \ell)$ be the radius in (16), with $p_{t,k}$ replaced by $p_{t,k,\ell}$ and $\varepsilon_{\theta, \text{eff}}$ replaced by $\varepsilon_{\omega, \text{eff}}$ below. Define

$$\text{UCB}_t^\dagger(k, \ell) = \widehat{r}_t^\dagger(k, \ell) + b_t^\dagger(k, \ell), \quad \text{LCB}_t^\dagger(k, \ell) = \widehat{r}_t^\dagger(k, \ell) - b_t^\dagger(k, \ell),$$

$$\text{UCB}_t^\dagger(k) = \max_{\ell \in \mathcal{L}_t(k)} \text{UCB}_t^\dagger(k, \ell), \quad \text{LCB}_t^\dagger(k) = \max_{\ell \in \mathcal{L}_t(k)} \text{LCB}_t^\dagger(k, \ell).$$

The surviving offsets and selected action are

$$\begin{aligned} \mathcal{K}_t^\dagger &= \left\{ k \in \mathcal{A}_t^\dagger : \text{UCB}_t^\dagger(k) \geq \max_{j \in \mathcal{A}_t^\dagger} \text{LCB}_t^\dagger(j) \right\}, \\ k_t &\in \arg \min_{k \in \mathcal{K}_t^\dagger} N_k(t), \quad \ell_t \in \arg \max_{\ell \in \mathcal{L}_t(k_t)} \text{UCB}_t^\dagger(k_t, \ell), \\ (p_t, \mu_t) &= (p_{t,k_t,\ell_t}, \mu_{\ell_t}). \end{aligned}$$

The grid-coverage requirement must also respect pairwise feasibility. Let $(p_t^{\dagger,*}, \mu_{\ell_t^{\dagger,*}})$ maximize $r^\dagger(x_t; p, \mu)$. At every Mode 0 round, assume there exists an offset k such that

$$\ell_t^{\dagger,*} \in \mathcal{L}_t(k), \quad \left| \delta_k - \left(p_t^{\dagger,*} - \phi_{t,\ell_t^{\dagger,*}}^\top \omega \right) \right| \leq \eta_\Delta.$$

As in Theorem 1, take $K = \mathcal{O}(T^{1/3})$ and $\eta_\Delta = \mathcal{O}(T^{-1/3})$. Feasibility at some other deadline, which only guarantees $k \in \mathcal{A}_t^\dagger$, is insufficient for the oracle comparison.

Informative deadlines and matured valuation data. After the timing gate passes, enter Mode 1 whenever

$$\max_{\ell \in [L]} \|\phi_{t,\ell}\|_{V_{\omega,t-1}^{-1}} > \eta_\omega.$$

Choose a maximizer ℓ_t , set $\mu_t = \mu_{\ell_t}$, and then draw $p_t \sim \text{Unif}[\underline{p}, \bar{p}]$. The deadline and ϕ_t are therefore measurable before the fresh price and customer shocks are drawn. Store

$$q_t = \widehat{\Psi}_{t-1}^c(x_t, p_t, \mu_t) \in [\pi_{\min}/2, 1].$$

At the offer's actual deadline maturity $t + \mu_t$, form

$$\widetilde{y}_t^{\dagger, \text{ipw}} = \underline{p} + R_p \frac{B_t}{q_t}, \quad R_p = \bar{p} - \underline{p},$$

without recomputing q_t . Estimates indexed by t include probes that have matured by calendar time t ; decisions in round t use the estimates indexed by $t - 1$. The augmented ridge data are

$$\begin{aligned} \mathcal{I}_t^\omega &:= \{s \leq t : i_s = 1, s + \mu_s \leq t\}, & N_t^\omega &:= |\mathcal{I}_t^\omega|, \\ V_{\omega,t} &:= \lambda I_{\widetilde{d}} + \sum_{s \in \mathcal{I}_t^\omega} \phi_s \phi_s^\top, \\ \widehat{\omega}_t &:= V_{\omega,t}^{-1} \sum_{s \in \mathcal{I}_t^\omega} \phi_s \widetilde{y}_s^{\dagger, \text{ipw}}. \end{aligned}$$

Both uncensored and censored probes contribute at maturity, since $B_s = 0$ when $w_s = 0$.

Confidence radius and thresholds. For a fixed horizon, a sufficient augmented valuation radius is

$$\beta_T^{\omega, \text{ipw}}(\zeta) := \frac{R_p}{\pi_{\min}} \sqrt{2 \left[\widetilde{d} \log \left(1 + \frac{TB_\phi^2}{\lambda \widetilde{d}} \right) + \log \frac{T}{\zeta} \right]} + \sqrt{\lambda} B_\omega.$$

The additional $\log T$ permits a union bound over fixed calendar times; with varying deadlines, matured probes need not form a prefix of the posting sequence. Set $\zeta_T = 1/(6T)$ and

$$\begin{aligned} \varepsilon_\omega &= \widetilde{d}^{2/3} \sqrt{\log T} T^{-1/3}, & \varepsilon_\psi &= (d')^{2/3} \sqrt{\log T} T^{-1/3}, \\ \eta_\omega &= \frac{\varepsilon_\omega}{\beta_T^{\omega, \text{ipw}}(\zeta_T)}, & \eta_h &= \frac{\varepsilon_\psi}{\max\{L_g, L_C\} \beta_T^\psi(\zeta_T)}, \\ \varepsilon_{\omega, \text{eff}} &= \varepsilon_\omega + C_{\omega\psi} \sqrt{\widetilde{d} \log T} \varepsilon_\psi. \end{aligned}$$

Here $C_{\omega\psi}$ absorbs the IPW constant $C_{\text{ipw}} = 2R_p/\pi_{\min}$ and the bounded-delay exploration constants, as $C_{\theta\psi}$ does in the base theorem. Take T large enough that $\varepsilon_\psi \leq \pi_{\min}/2$.

Proof adaptations. Let $\mathcal{G}_s$ be the information available immediately before the uniform price in a Mode 1 probe, including the selected deadline. Conditional separation and the uniform support condition give

$$\begin{aligned}\mathbb{E}\left[\underline{p} + R_p \frac{B_s}{\Psi(x_s, p_s, \mu_s)} \middle| \mathcal{G}_s\right] &= \underline{p} + \int_{\underline{p}}^{\bar{p}} D(p - \phi_s^\top \omega) dp \\ &= \phi_s^\top \omega.\end{aligned}$$

On the timing-confidence event, the unchanged timing gate gives $|q_s - \Psi(x_s, p_s, \mu_s)| \leq \varepsilon_\psi$. Clipping then bounds the pseudo-label's conditional bias by $C_{\text{ipw}}\varepsilon_\psi$, and its centered noise is conditionally $R_p/\pi_{\min}$ -sub-Gaussian, as in Lemma 1. For each fixed calendar time t , the inclusion condition $s + \mu_s \leq t$ is known before the probe's price and shocks. Applying the ridge concentration argument to these predictable selections, then taking a union bound over $t \leq T$, gives

$$|\phi^\top(\widehat{\omega}_t - \omega)| \leq \left[\beta_T^{\omega, \text{ipw}}(\zeta_T) + C_{\text{ipw}}\varepsilon_\psi \sqrt{N_t^\omega} \right] \|\phi\|_{V_{\omega, t}^{-1}}$$

on the joint confidence event. This argument uses predictability and the stated sequential shock conditions; it does not require the adaptively chosen deadlines to be independent of past data.

Every selected Mode 1 feature has leverage above η_ω , and its feedback arrives within $\bar{\mu}$ periods. The delayed elliptical-potential counting argument of Lemma EC.1 consequently gives $|G_1| = \tilde{O}(\bar{d}/\eta_\omega^2)$ and $N_t^\omega \eta_\omega^2 = O(\bar{d} \log T)$, with the same bounded-delay dependence. At a Mode 0 round, the maximum in the valuation gate therefore ensures

$$\max_{\ell \in [L]} |\phi_{t, \ell}^\top(\widehat{\omega}_{t-1} - \omega)| \leq \varepsilon_{\omega, \text{eff}}.$$

This controls every candidate deadline, including deadlines different from those used in valuation exploration.

For a feasible Mode 0 pair,

$$p_{t, k, \ell} - \phi_{t, \ell}^\top \omega = \delta_k + \phi_{t, \ell}^\top(\widehat{\omega}_{t-1} - \omega).$$

Hence the Lipschitz property of D bounds its demand discrepancy from $D(\delta_k)$ by $L_D \varepsilon_{\omega, \text{eff}}$, uniformly over the feasible pairs. Conditional separation still factors each observed purchase probability into this demand and its own deadline-meeting probability. The exposure-normalized argument in Lemma 3 thus applies to the same pooled scalar $D(\delta_k)$, with augmented valuation error and unchanged timing and pending-offer terms.

Finally, define the current grid value

$$\bar{v}_t^\dagger(k) := \max_{\ell \in \mathcal{L}_t(k)} r^\dagger(x_t; p_{t,k,\ell}, \mu_\ell).$$

The coverage condition provides a feasible price within $\eta_\Delta + \varepsilon_{\omega, \text{eff}}$ of $p_t^{\dagger,*}$ at the oracle's own deadline. At a fixed deadline the augmented index is constant in price, so the base price-Lipschitz argument bounds the discretization loss by $O(T\eta_\Delta + T\varepsilon_{\omega, \text{eff}})$. Pairwise confidence bounds preserve the maximizing grid offset under elimination, and the same selected-offset counting argument yields

$$\tilde{O}(K/\alpha + T\alpha + T\varepsilon_{\omega, \text{eff}} + T\varepsilon_\psi + K\bar{\mu})$$

for the remaining Mode 0 regret. With $\alpha = T^{-1/3}$ and the thresholds above, adding the valuation- and timing-exploration terms gives

$$\text{Regret}_T^\dagger = \tilde{O}(T^{2/3}) \quad \text{with probability at least } 1 - O(T^{-1}).$$

The bound holds under the augmented regularity, support, separation, and oracle-deadline coverage conditions just stated, for fixed $\tilde{d}, d'$, deadline menu, and operating-domain constants.

EC.7. Timing-Channel Urgency

The baseline model allows deadlines to affect realized conversion through the timing channel. In particular, an urgency effect need not be modeled as a direct shift in the customer's intrinsic valuation of the product. Even when the latent purchase probability $D(p - x^\top \theta)$ is fixed for a given (x, p) , shorter deadlines can increase realized conversion if they accelerate the customer's consideration process enough to offset the shorter response window.

To illustrate this point, fix x and p , and write

$$D_0 := D(p - x^\top \theta).$$

For exposition, treat the deadline μ as continuous and positive; the same comparison applies on any finite deadline menu. Consider an exponential consideration-time model in which the hazard depends on the posted deadline:

$$\rho_\alpha(\mu) = \rho_0 \left(\frac{\bar{\mu}}{\mu} \right)^\alpha,$$

where $\rho_0 > 0$ is a baseline hazard and $\alpha \geq 0$ controls the strength of urgency. The probability that the decision arrives before the deadline is

$$\Psi_\alpha(\mu) = 1 - \exp\{-\mu\rho_\alpha(\mu)\} = 1 - \exp\{-\rho_0\bar{\mu}^\alpha\mu^{1-\alpha}\}.$$

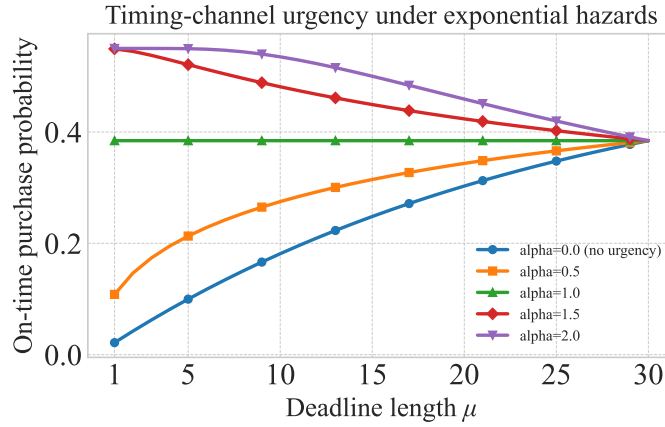


Figure EC.1 Timing-channel urgency under an exponential-hazard model. The latent purchase probability $D(p - x^\top \theta)$ is fixed across deadlines, while the consideration hazard is $\rho_\alpha(\mu) = \rho_0(\bar{\mu}/\mu)^\alpha$. For $\alpha > 1$ and positive latent purchase propensity, shorter deadlines increase the hazard enough that the probability of an on-time purchase increases as the deadline is shortened. Thus urgency can be represented through the timing channel without directly shifting intrinsic valuation.

Therefore the probability of an on-time purchase is

$$S_\alpha(\mu) = D_0 \Psi_\alpha(\mu) = D_0 \left[1 - \exp \left\{ -\rho_0 \bar{\mu}^\alpha \mu^{1-\alpha} \right\} \right].$$

Differentiating with respect to μ gives

$$\frac{dS_\alpha(\mu)}{d\mu} = D_0 \exp \left\{ -\rho_0 \bar{\mu}^\alpha \mu^{1-\alpha} \right\} \rho_0 \bar{\mu}^\alpha (1 - \alpha) \mu^{-\alpha}.$$

For $D_0 > 0$, the sign of $dS_\alpha(\mu)/d\mu$ is determined by $1 - \alpha$; if $D_0 = 0$, conversion is zero at every deadline. For positive D_0 , when $0 \leq \alpha < 1$, longer deadlines increase the probability of an on-time purchase. When $\alpha = 1$, the shorter response window is exactly offset by the higher consideration hazard, so $S_\alpha(\mu)$ is constant in μ . When $\alpha > 1$, the acceleration effect dominates: $S_\alpha(\mu)$ decreases as μ increases, so shortening the deadline increases the probability of an on-time purchase.

The same family also permits an explicit comparison of open-offer costs. For the integer deadlines in the main model, let $\tilde{d}_\mu \sim \text{Exp}(\rho_\alpha(\mu))$ and $d_\mu = \lceil \tilde{d}_\mu \rceil$. The expected open time is

$$C_\alpha(\mu) = \mathbb{E}[\min\{d_\mu, \mu\}] = \sum_{j=0}^{\mu-1} \mathbb{P}(d_\mu > j) = \sum_{j=0}^{\mu-1} \exp\{-j\rho_\alpha(\mu)\}.$$

For $\alpha \geq 0$, the hazard is nonincreasing in μ . Increasing the integer deadline therefore weakly increases every existing summand and adds strictly positive terms, so $C_\alpha(\mu)$ is strictly increasing. When $\alpha \geq 1$, conversion is nonincreasing in μ while expected open time is strictly increasing. Consequently, at

the fixed price p and any positive holding-cost rate, the shortest feasible deadline strictly maximizes expected profit within this family. For $\alpha > 1$ and $D_0 > 0$, shortening the deadline improves both conversion and holding costs; for $\alpha = 1$, conversion is unchanged. The conclusion extends to the joint price–deadline optimum if the same urgency condition holds at every feasible price for the given context.

This example clarifies the role of the timing model in the baseline analysis. The deadline can affect whether and when customers resolve the offer through $\Psi(x, p, \mu)$, while the intrinsic buy–reject margin remains governed by the price offset $p - x^\top \theta$. The deadline-augmented formulation in Section 6 covers the complementary case in which deadlines also directly shift purchase propensity.